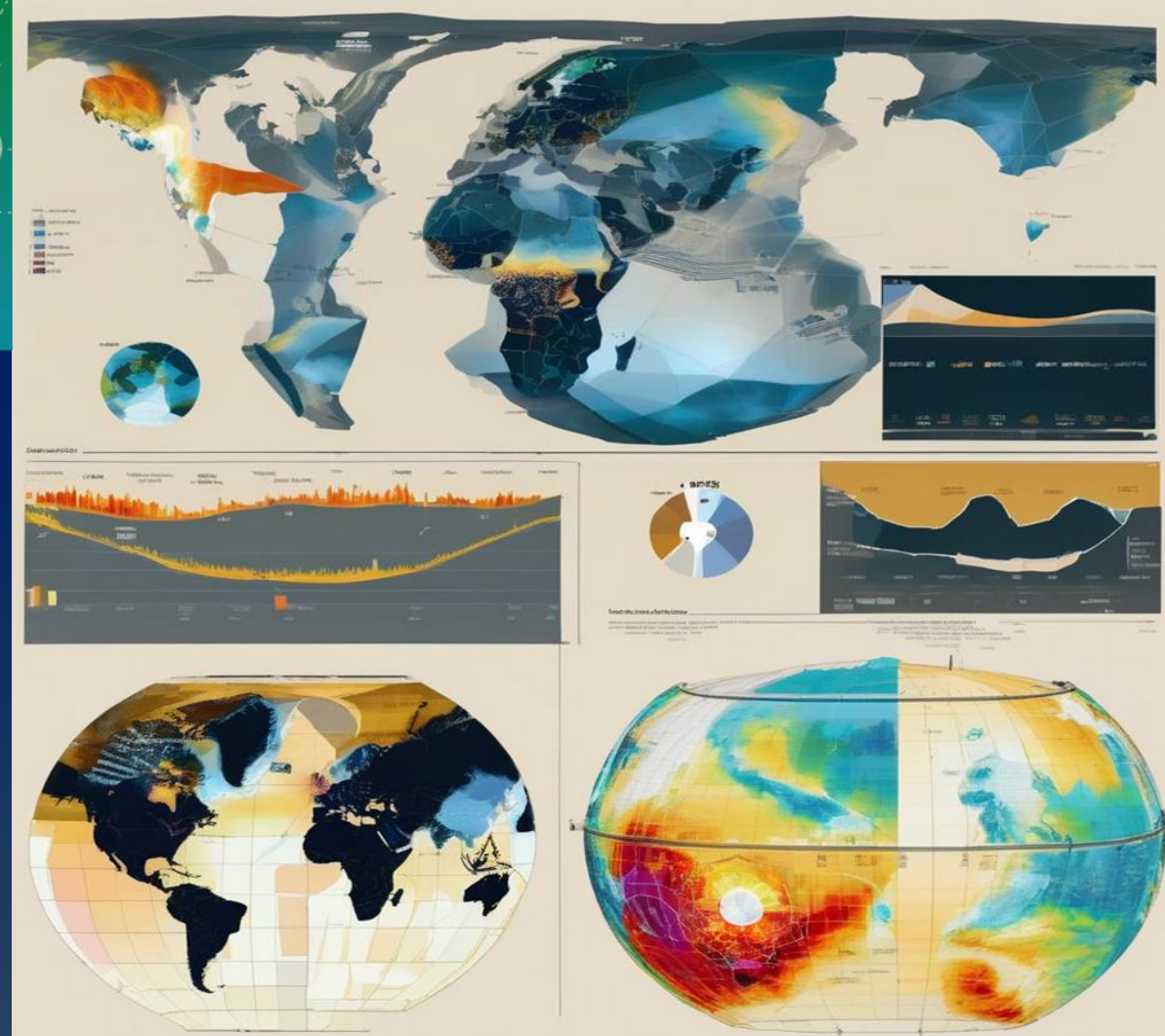




Aden (Yemen), 2 Dec - 4 Dec., 2025

Session III. Introduction to CAVA



Andrea Setti: andrea.setti@fao.com

Climate Risk Specialist

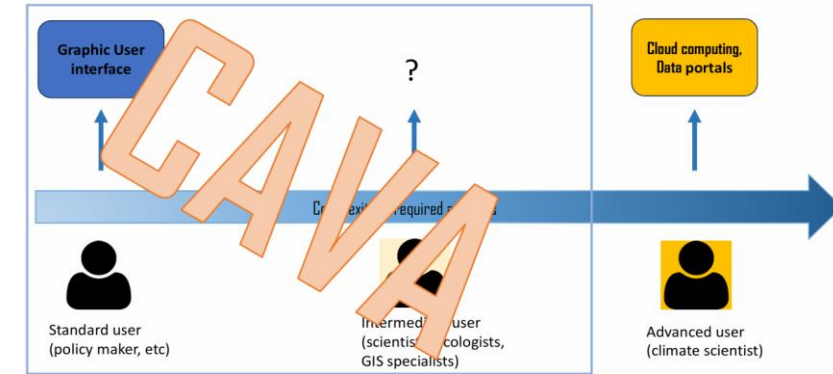
**Office of Climate Change, Biodiversity and Environment
(OCB, FAO)**

Table of contents: Session III

1. Introduction to CAVA
2. Interpretation of maps

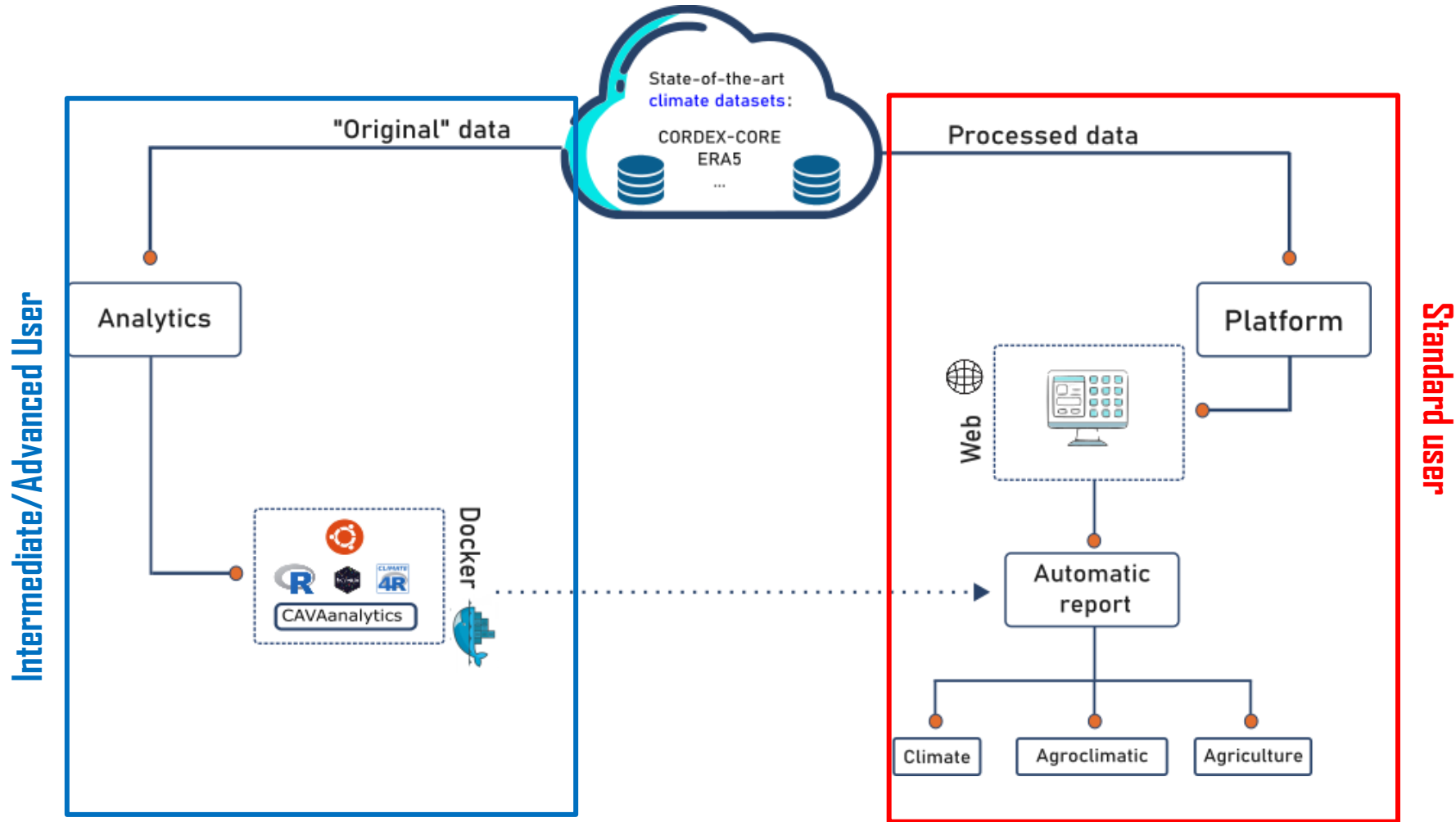
1. Introduction to CAVA

- **CAVA (Climate and Agriculture Risk Visualization and Assessment)** is an agriculture-focused climate information approach that aims to democratize access to and use of climate and impact information for the delivery of long-term climate services.
- The CAVA platform allows users to view a number of pre-calculated climate and agro-climatic indicators. In addition, users can receive comprehensive reports summarizing agro-climatology tailored to their specific regions of interest via email.
- For advanced users seeking in-depth analytical capabilities, CAVA Analytics enables customized analyses and the use of CORDEX-CORE data for more precise applications.



IT Framework: CAVA platform vs. CAVA Analytics

Simplified access to climate models, their processing
and visualization



Graphical user interface with some advanced
computing capabilities

The automated report allows users to benefit from the capabilities of CAVA Analytics.

Create an account

CAVA: Access to the platform



Food and Agriculture Organization
of the United Nations



Universidad
de Cantabria

Sign In 

CAVA: CLIMATE AND AGRICULTURE RISK VISUALIZATION AND ASSESSMENT

Climate and Agriculture Risk Visualization and Assessment (CAVA) is an approach to climate information with a focus on agriculture. Its goal is to provide long-term, climate services by democratising access and simplifying the use of past and future climate and impact data.

CAVA offers two main components:

1. CAVA Platform: a freely accessible and user-friendly platform for climate and impact information
2. CAVAnalytics: dedicated R package providing advanced analyses and harmonized access to raw data used by the CAVA Platform

It is developed through a collaboration between the Food and Agriculture Organization of the United Nations and the University of Cantabria

You can use all features for free

Sign In 

Sign Up 

Step 1: click on SIGN UP

<https://cavaplatfrom.com/>

CAVA: Create an account

CREATE AN ACCOUNT TO ACCESS CAVA

<https://cavaplatform.com/register>



The registration form is titled "Food and Agriculture Organization of the United Nations" and features the FAO logo. It contains four input fields: "Email*", "Username*", "Password*", and "Confirm password*", each outlined with a red border. At the bottom, there is a link "I already have the registration token" and a blue "Register >" button, also outlined with a red border.

Step 1: Enter your email address

Step 2: Create a username

Step 3: Create a password

Step 4: Confirm the password

Step 5: Click on REGISTER

CAVA: activation of the account

ACTIVATE YOUR ACCOUNT



Food and Agriculture Organization
of the United Nations

Enter the code you received in your email and your username. If you have not received any email, please check your spam folder or contact your system administrator.

Username*

Code*

Activate account ✓

Step 1: Enter your username

Step 2: Enter the code you received by email (check your spam folder if you can't find it)

Step 3: Click on **ACTIVATE ACCOUNT**

CAVA: Start CAVA

WELCOME TO CAVA

Climate and Agriculture risk Visualization and Assessment



Food and Agriculture Organization
of the United Nations

Username*

JAlvar-563

Password*

.....

[Forgot your password?](#)

Register

Login

Step 1: Enter your username

Step 2: Enter your password

Step 3: Click on **LOGIN**

CAVA: Initial interface

← → ↻ cavaplatfrom.com/map

CAVA: CLIMATE AND AGRICULTURE RISK VISUALIZATION AND ASSESSMENT

Profile About Glossary

Q Only selected Sync layers

- > Climate variables 0 / 20
- > Agro-climatic 0 / 95
- > Agriculture

Atlantic Ocean

Indian Ocean

Pacific Ocean

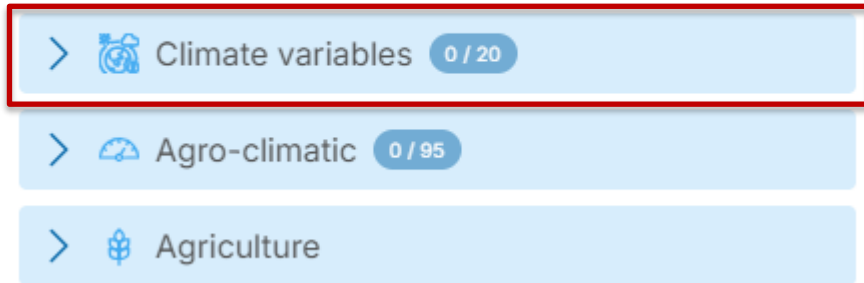
Map showing world countries and oceans.

This option is not yet available

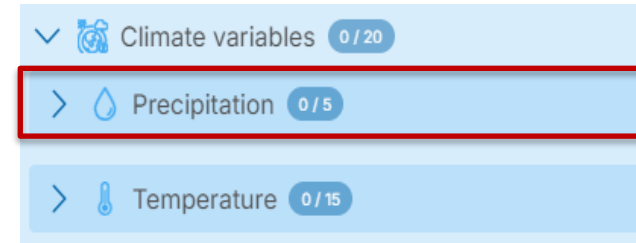
Visualizing: Historical rainfall and temperatures

CAVA: Historical rainfall

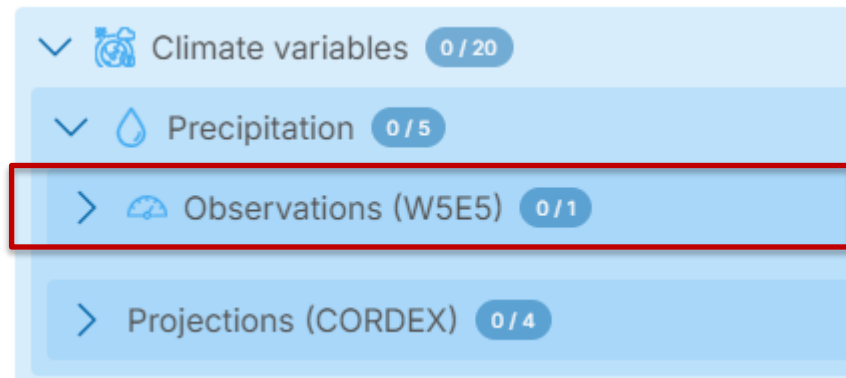
 Only selected ☐ Sync layers 



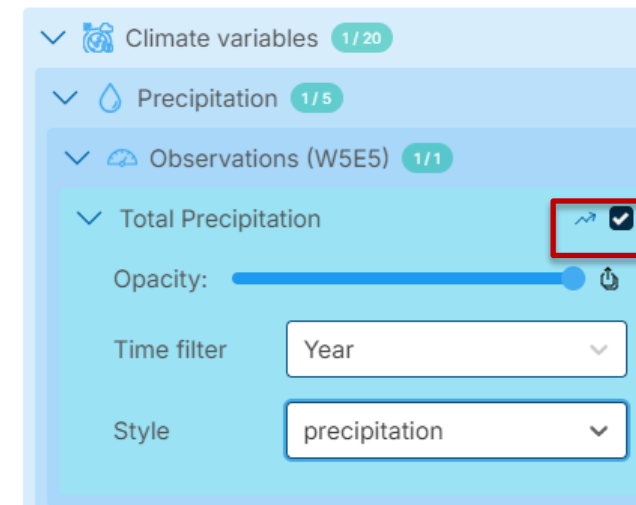
Step 1: Click on
**CLIMATE
VARIABLES**



Step 2: Click on
PRECIPITATION



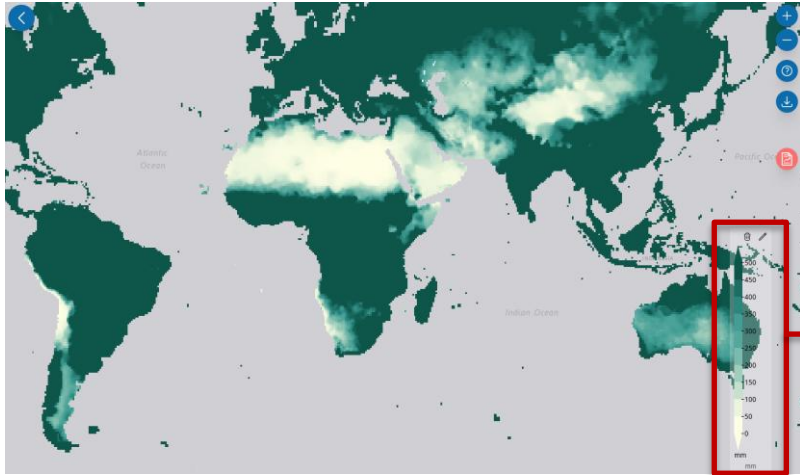
Step 3: Click on
OBSERVATIONS



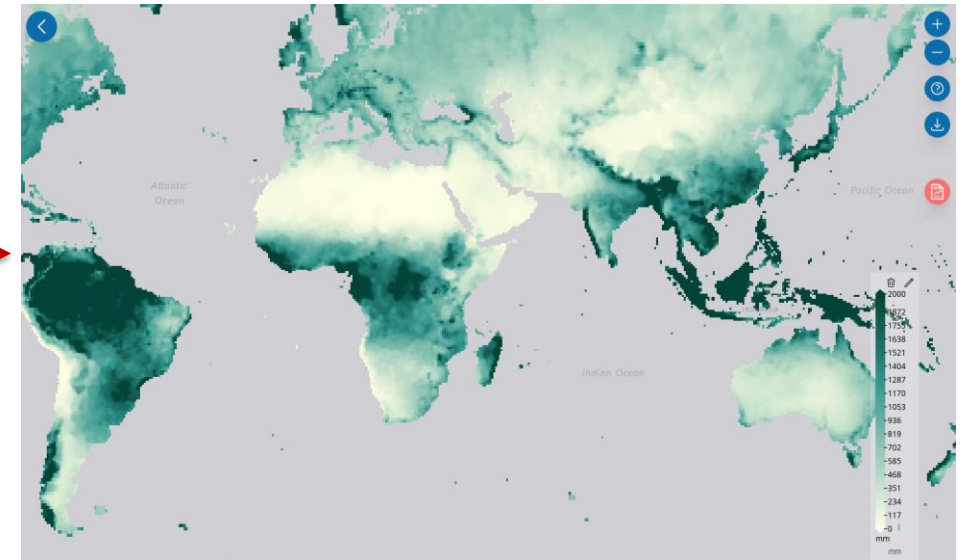
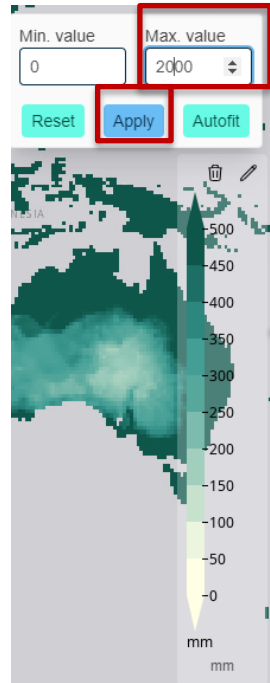
Step 4: Make sure
the layer is selected

CAVA: Historical preccipitations

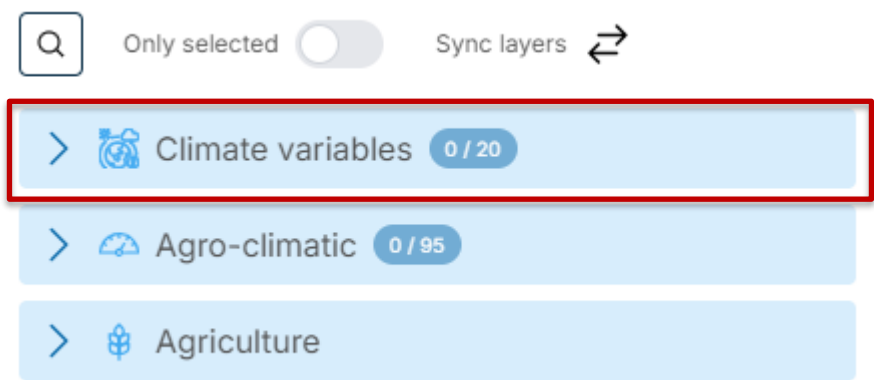
Step 2: Change the maximum value
(2000) and click **APPLY**



Step 1: Change
the legend



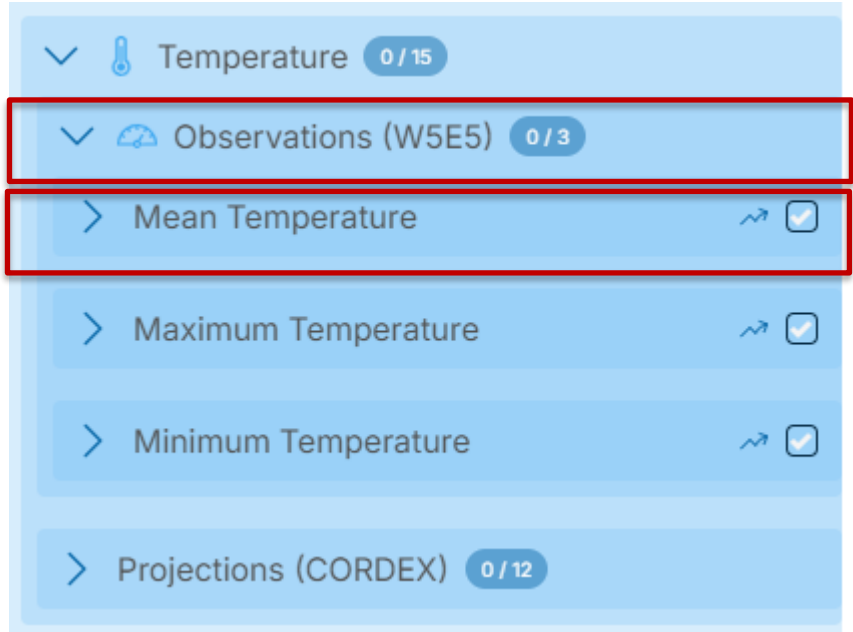
CAVA: Historical Temperatures



Step 1: Click on
CLIMATE VARIABLES

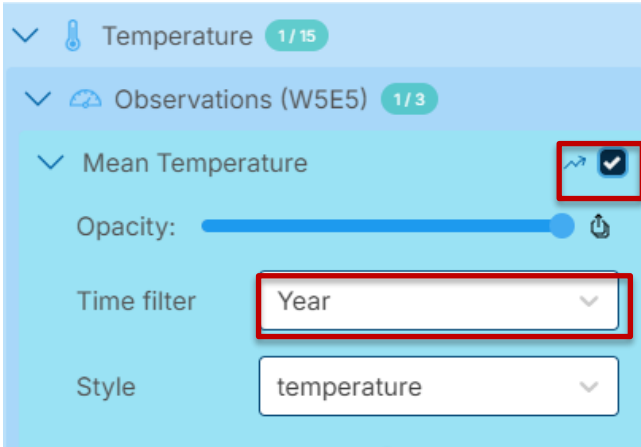


Step 2: Click on
TEMPERATURE



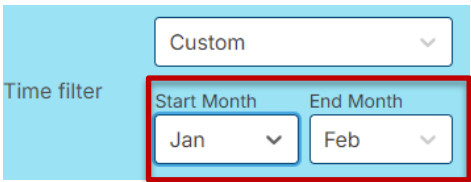
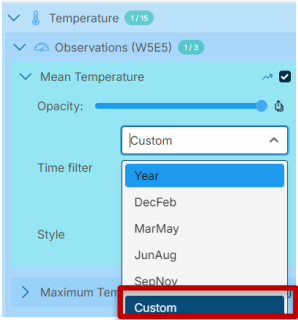
Step 3: Click on
OBSERVATIONS

Step 4: Click on
MEAN TEMPERATURE



Step 5: Ensure the
layer is selected

You can select
different periods
(e.g. January-
February)

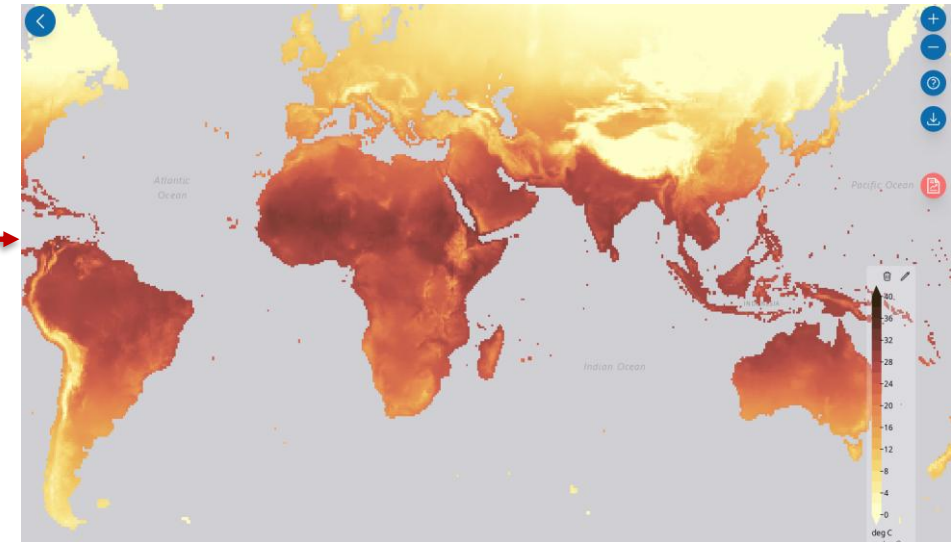
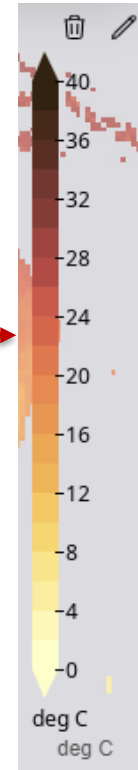
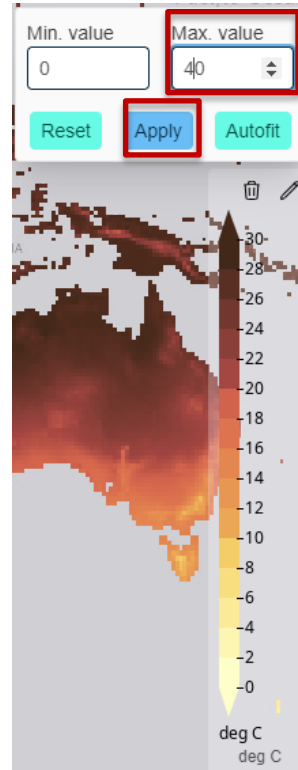


CAVA: Historical Temperatures

Step 2: Change the maximum value
(40) and click **APPLY**



Step 1: Change
the legend



Agro-climatic indices

CAVA: Agro-climatic indices (precipitation)

Q Only selected Sync layers

> Climate variables 0 / 20

> Agro-climatic 0 / 95

> Agriculture

Step 1: Click on
AGROCLIMATIC

> Average length of dry spells

> Maximum length of dry spell

> Number of dry spells (10 or more dry days)

> Number of dry spells (5 or more dry days)

> Number of wet spells (5 or more days with precip >= 10mm)

> Number of wet spells (5 or more days with precip >= 20mm)

> Number of rainy days (>=1mm)

> Rainy day (>=1mm) average precipitation

> Maximum consecutive 5-day precipitation

> Number of high rainfall days (>= 50mm)

Step 4: Select
AVERAGE LENGTH OF DRY SPELLS

There are 13 agro-climatic indices related to rainfall that are prepared by default in CAVA.

> Agro-climatic 0 / 95

> Precipitation 0 / 65

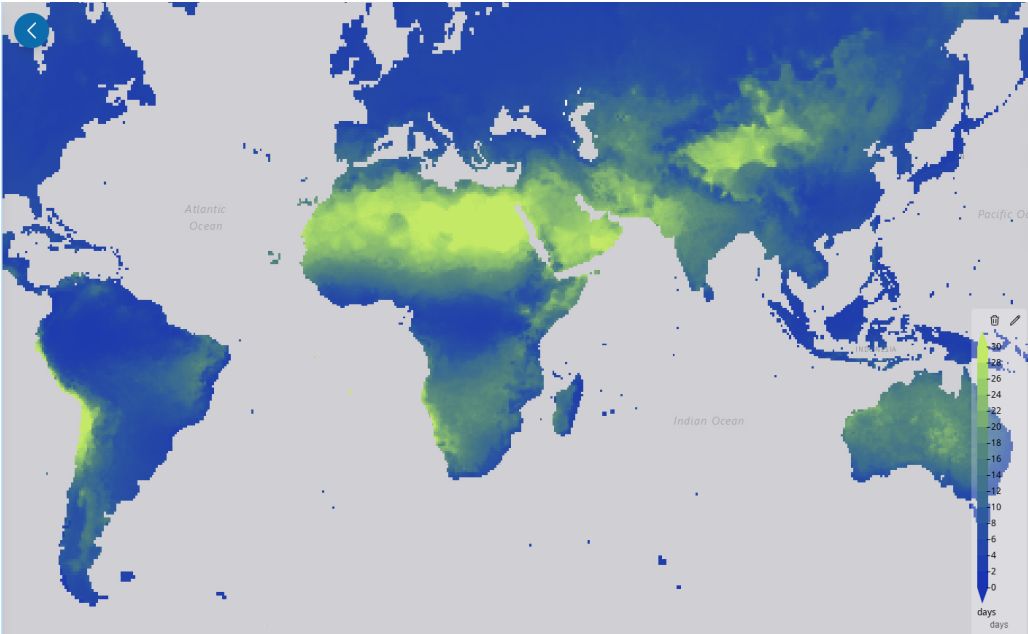
> Observations 0 / 13

> Projections (CORDEX) 0 / 52

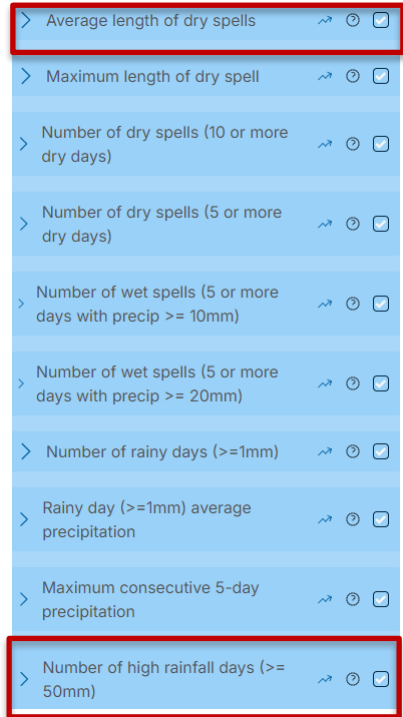
> Temperature 0 / 30

Step 2: Click on
PRECIPITATION

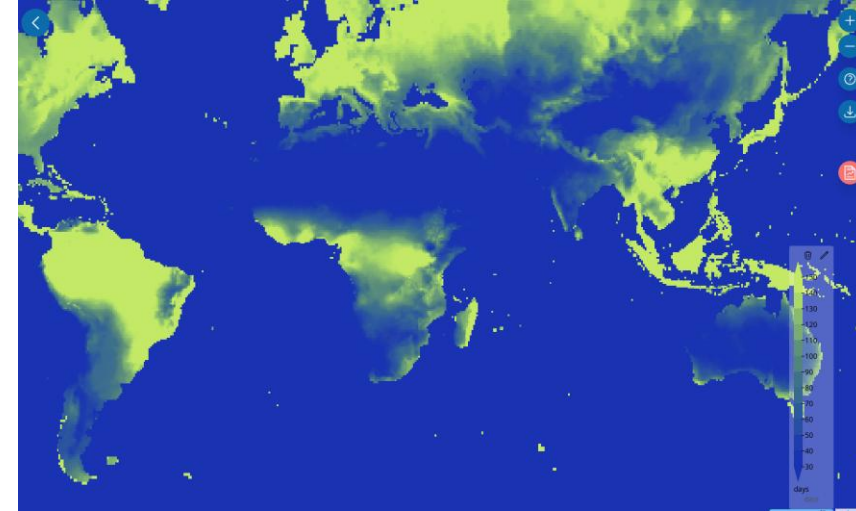
Step 3: Click on
OBSERVATIONS



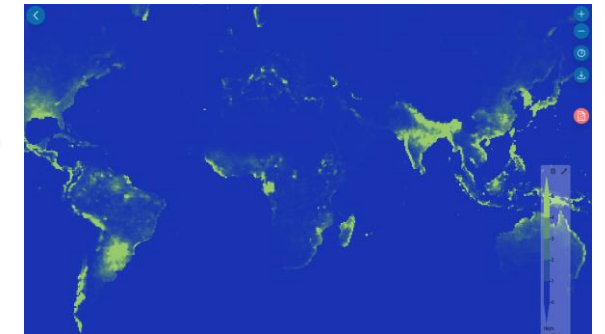
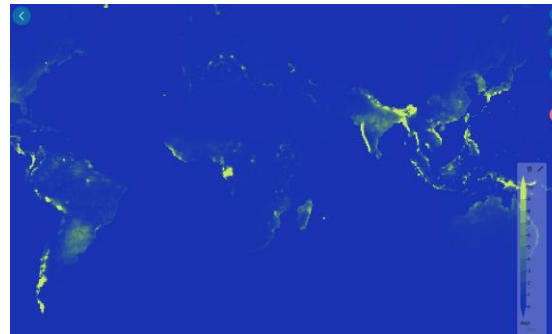
CAVA: Agro-climatic indices (precipitation)



**Step 1: Select
NUMBER OF RAINY DAYS
($>1\text{mm}$)**



**Step 2: Select
NUMBER OF HEAVY
RAINFALL DAYS ($>50\text{mm}$)**



You can change the legend to 5

CAVA: Agro-climatic indices (temperature)

Q Only selected Sync layers

> Climate variables 0 / 20

> Agro-climatic 0 / 95

> Agriculture

Step 1: Click on
AGROCLIMATIC

Temperature 0 / 30

Observations 0 / 6

> Number of heat waves (3 days)

> Number of heat waves (6 days)

> Number of cold Tmin days (≤ 0 degC)

> Number of hot Tmin days (≥ 20 degC)

> Number of hot Tmax days (≥ 30 degC)

> Number of hot Tmax days (≥ 35 degC)

Step 4: Select
NUMBER OF HEATWAVES

There are 6 agro-climatic
indices linked to
temperatures that are set
by default in CAVA

Agro-climatic 0 / 95

Precipitation 0 / 65

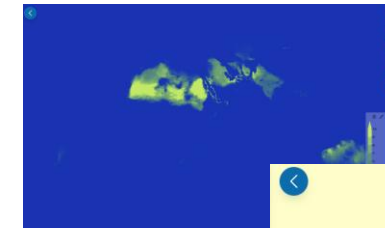
Temperature 0 / 30

Observations 0 / 6

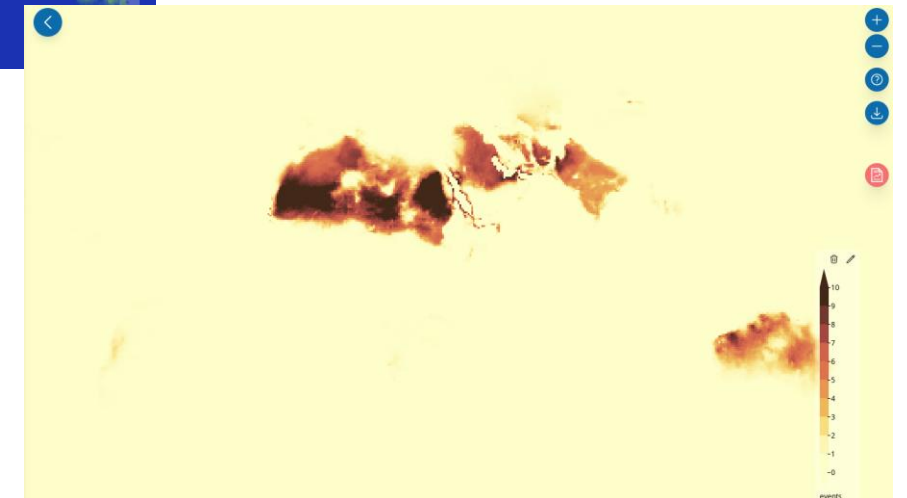
Projections (CORDEX) 0 / 24

Step 2: Click on
TEMPERATURE

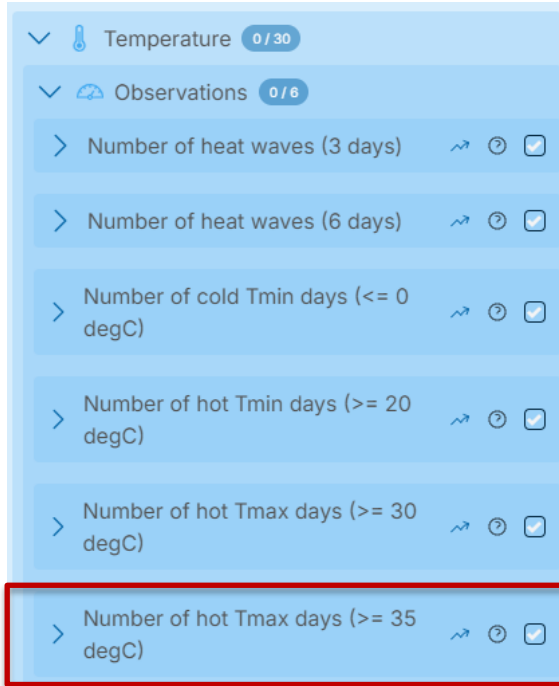
Step 3: Click on
OBSERVATIONS



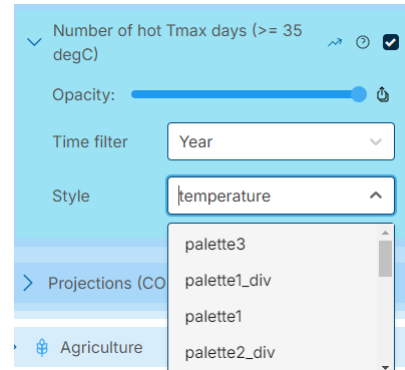
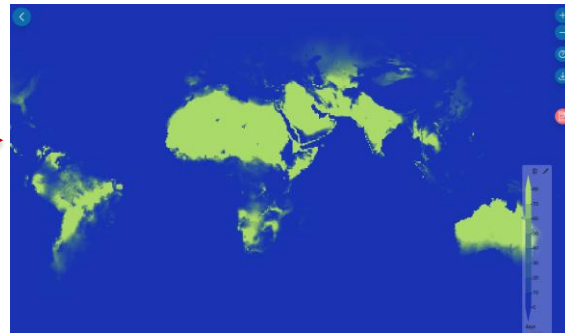
Change the color palette for
better visualization



CAVA: Agro-climatic indices (temperature)



**Step 1: Select
NUMBER OF HOT TMAX
DAYS (>35°C)**



**Step 2: Change the color
palette for better
visualization**



Visualizing future precipitation and temperature

CAVA: Initial Interface

← → ↻ cavaplatfrom.com/map

CAVA: CLIMATE AND AGRICULTURE RISK VISUALIZATION AND ASSESSMENT

Profile About Glossary

Q Only selected Sync layers

> Climate variables 0 / 20

> Agro-climatic 0 / 95

> Agriculture

Atlantic Ocean

Indian Ocean

Pacific Ocean

Map showing world countries and oceans. The 'Agriculture' option in the left sidebar is highlighted with a red box.

This option is not yet available

CAVA: Future annual precipitation (total)

 Only selected ☐ Sync layers 

>

 Climate variables 0 / 20

>

 Agro-climatic 0 / 95

>

 Agriculture

Step 1: Click on
CLIMATE VARIABLES

✓  Climate variables 0 / 20

>

 Precipitation 0 / 5

>

 Temperature 0 / 15

Step 2: Click on
PRECIPITATION

Here you will see the different
climate change scenarios

✓  Climate variables 0 / 20

✓  Precipitation 0 / 5

>

 Observations (W5E5) 0 / 1

>

 Projections (CORDEX) 0 / 4

Step 3: Click on
**PROJECTIONS
(CORDEX)**

✓ Projections (CORDEX) 0 / 4

>

RCP 2.6 0 / 2

✓ RCP 8.5 0 / 2

>

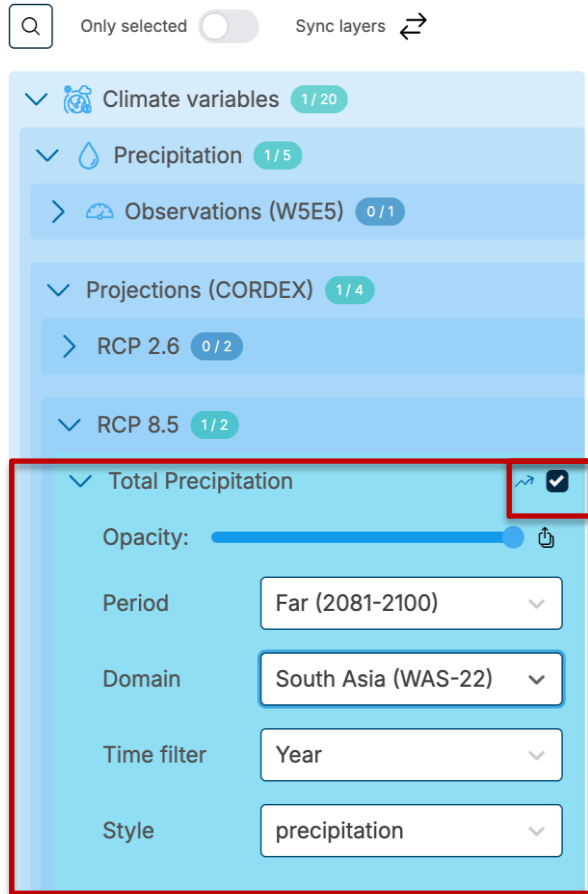
Total Precipitation  ☒

>

Total Precipitation Relative Anomaly ☒

Step 4: Click on **RCP 8.5**
Step 5: Click on **TOTAL
PRECIPITATION**

CAVA: Future annual precipitation (total)



Step 6: Ensure the layer is selected

Step 7: Choose the **period of interest** in the future (e.g., 2081-2100)

Step 8: Choose the **area of interest** (WAS-22)

Step 9: Choose the **period of the year of interest** (YEAR)

Step 10: Choose the **PRECIPITATION** color style

CAVA: CORDEX-CORE domains

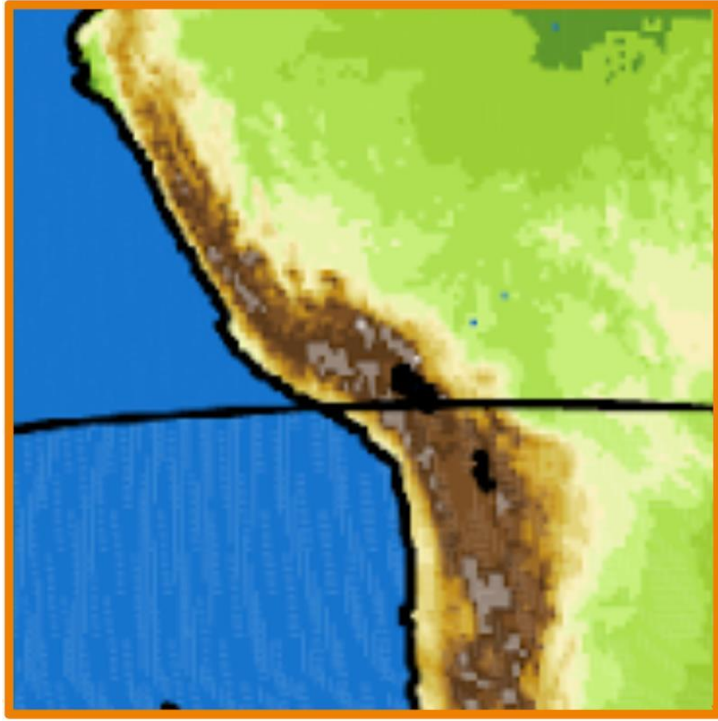


Figure 1b: Orographically structured area at CORDEX-CORE resolution of 0.22° .

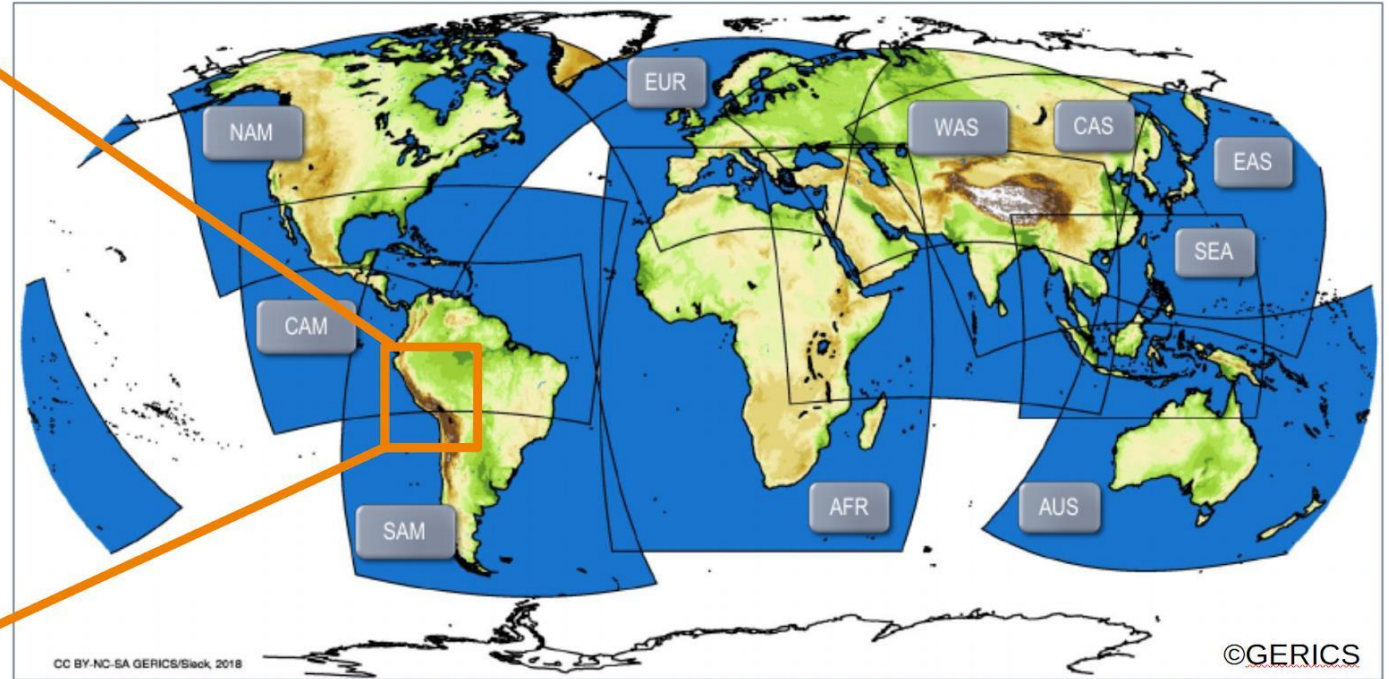
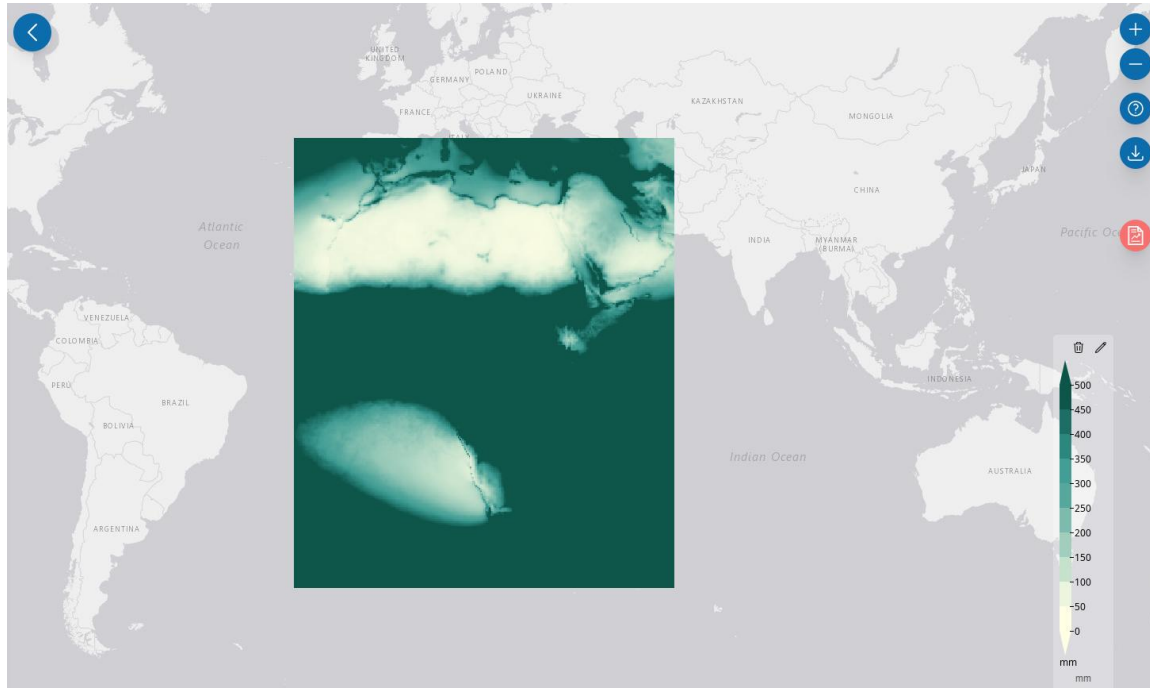


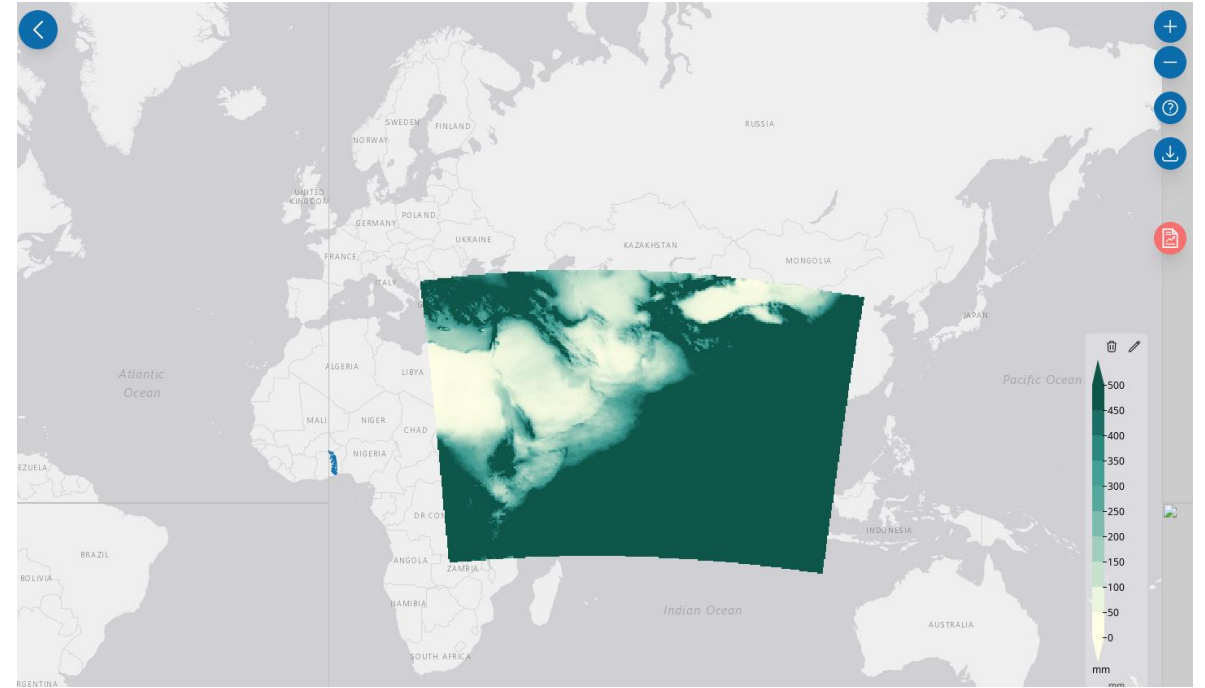
Figure 1a: CORDEX-CORE model domains used in the simulations. From top-left to bottom right: North America (**NAM**), Central America (**CAM**), South America (**SAM**), Europe (**EUR**), Africa (**AFR**), South Asia (**WAS**), East Asia (**EAS**), Southeast Asia (**SEA**), and Australasia (**AUS**). The domain Central Asia (**CAS**) is not officially included in CORDEX-CORE.

CAVA: CORDEX-CORE domains

AFR-22 = AFRICA



WAS-22 = SOUTH ASIA



CAVA: Future Precipitation (Anomaly)

✓ Precipitation 1/5

> Observations (W5E5) 0/1

✓ Projections (CORDEX) 1/4

> RCP 2.6 0/2

✓ RCP 8.5 1/2

> Total Precipitation

✓ Total Precipitation Relative Anomaly ☒

Opacity:

Period: Far (2081-2100) ▼

Domain: South Asia (WAS-22) ▼

Time filter: Year ▼

Baseline: AR5 (1986-2005) ▼

Style: precipitation_div ▼

Step 1: Select **TOTAL ANOMALY RELATIVE PRECIPITATION**

Step 2: Choose the **period of interest in the future** (e.g. 2081-2100)

Step 3: Choose the **Area of Interest** (WAS-22)

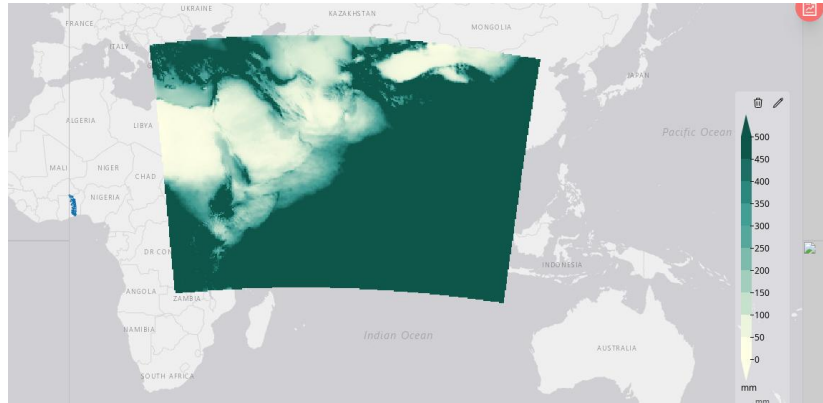
Step 4: Choose the **period of the year of interest** (YEAR)

Step 5: Choose the **period for comparing the simulations** (AR5 1986-2005)

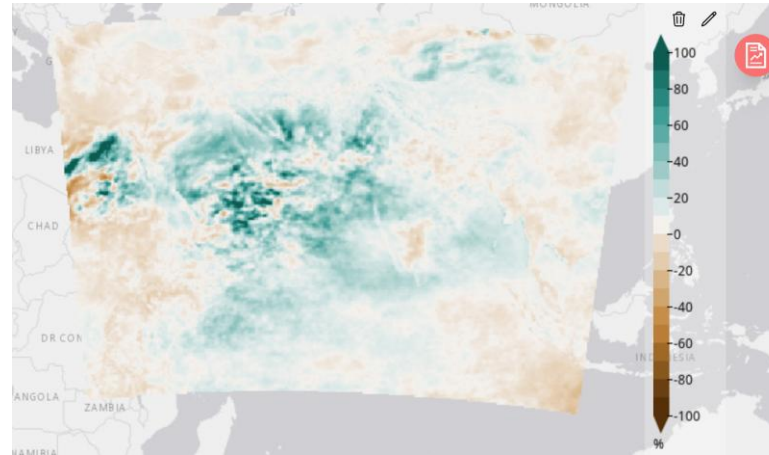
Step 6: Choose the **Style** (PRECIPITATION_DIV)

CAVA: Future Precipitation (Total vs. Anomaly)

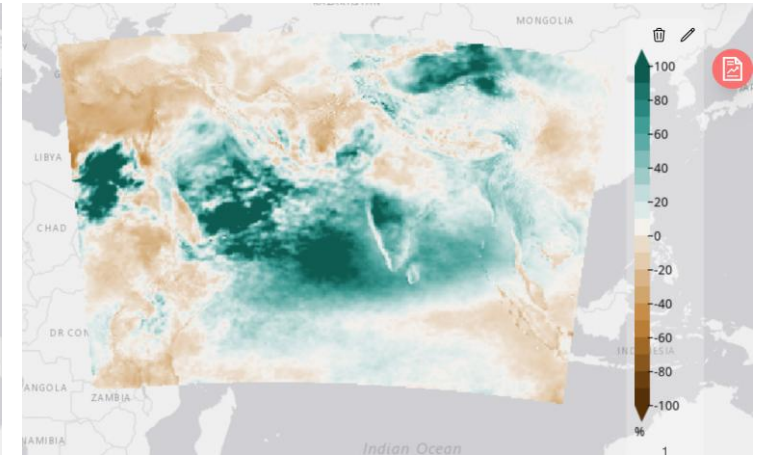
TOTAL PRECIPITATION (mm)



TOTAL PRECIPITATION (% anomaly)



RCP 2.6
(2080-2099 vs. 1986-2005)



RCP 8.5
(2080-2099 vs. 1986-2005)

CAVA: Future Average Temperatures

✓ Climate variables 0 / 20

✓ Precipitation 0 / 5

> Observations (W5E5) 0 / 1

> Projections (CORDEX) 0 / 4

✓ Temperature 0 / 15

> Observations (W5E5) 0 / 3

✓ Projections (CORDEX) 0 / 12

> RCP 2.6 0 / 6

✓ RCP 8.5 0 / 6

> Mean Temperature ~ ↗ ☒

> Mean Temperature Anomaly ☒

> Maximum Temperature ~ ↗ ☒

> Maximum Temperature Anomaly ☒

> Minimum Temperature ~ ↗ ☒

> Minimum Temperature Anomaly ☒

Future AVERAGE TEMPERATURES

Step 2: Click on TEMPERATURE

Step 3: Click on PROJECTIONS (CORDEX)

Step 4: Click on RCP 8.5

Step 5: Click on MEAN TEMPERATURE

CAVA: Future Average Temperatures

✓ Mean Temperature Anomaly ✓

Opacity:

Period: Far (2081-2100) ▼

Domain: South Asia (WAS-22) ▼

Time filter: Year ▼

Baseline: AR5 (1986-2005) ▼

Style: temperature_div ▼

Step 1: Choose the period of interest in the future (**e.g., 2081-2100**)

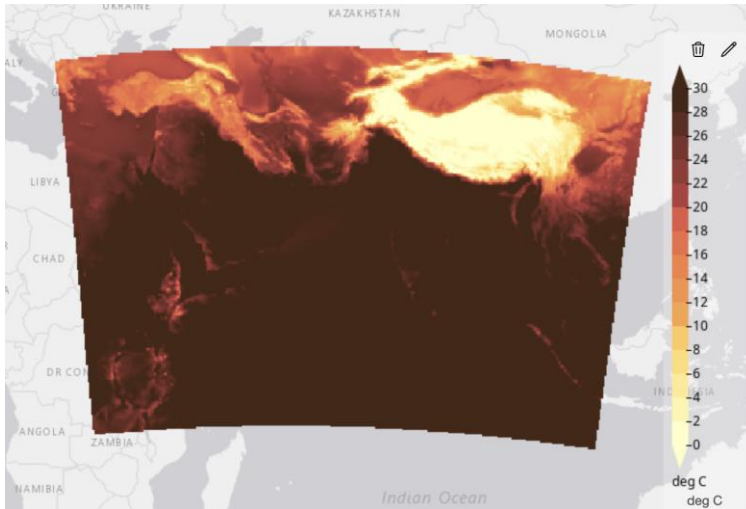
Step 2: Choose the area of interest (**WAS-22**)

Step 3: Choose the period of the year of interest (**YEAR**)

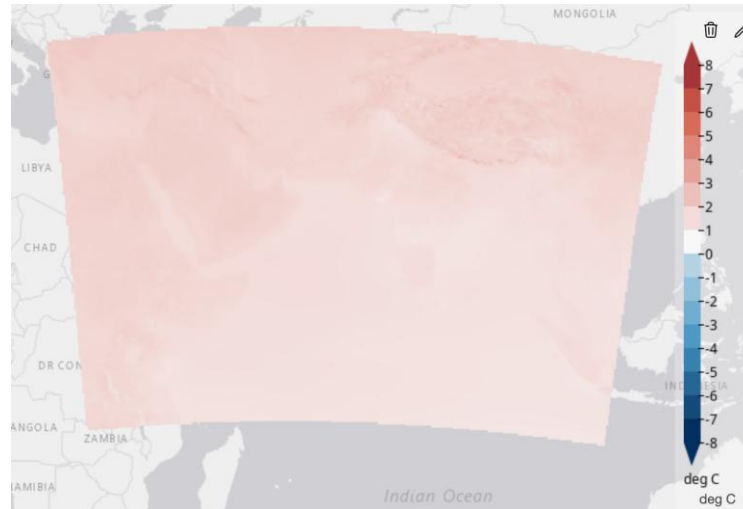
Step 4: Choose the style (**TEMPERATURE_DIV**)

CAVA: Future Average Temperatures (average vs. anomaly)

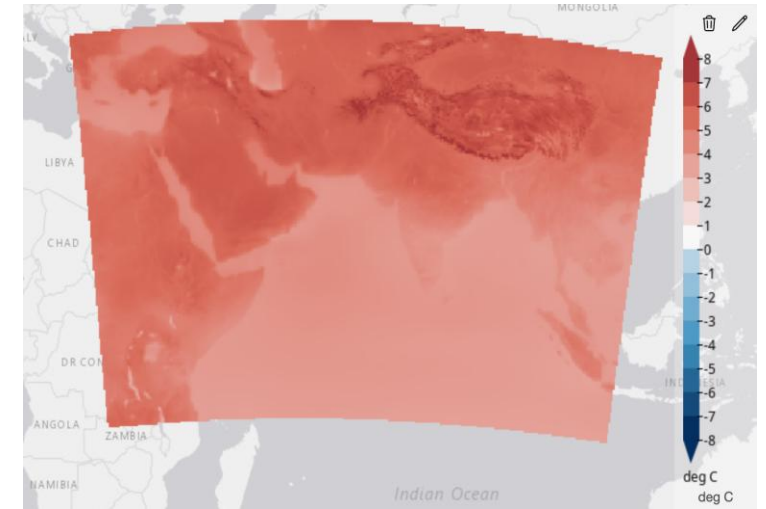
AVERAGE TEMPERATURE (°C)



TEMPERATURE (anomaly in °C)



RCP 2.6
(2080-2099 vs. 1986-2005)



RCP 8.5
(2080-2099 vs. 1986-2005)

Other information

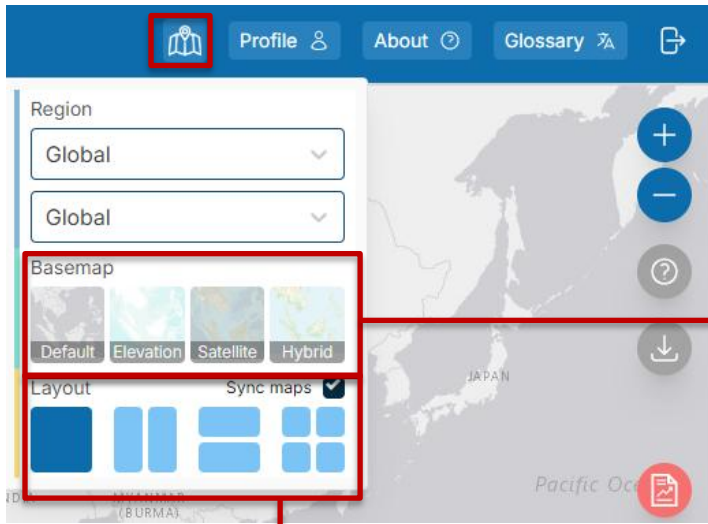
CAVA: Initial Interface

The screenshot displays the CAVA (Climate and Agriculture Risk Visualization and Assessment) web application interface. The browser address bar shows the URL `cavaplatform.com/map`. The header bar contains the CAVA logo and the text "CAVA: CLIMATE AND AGRICULTURE RISK VISUALIZATION AND ASSESSMENT". On the right side of the header, there are links for "Profile", "About", and "Glossary".

On the left side, there is a sidebar with a search icon and a "Only selected" toggle. Below this, there are three expandable menu items: "Climate variables" (0 / 20), "Agro-climatic" (0 / 95), and "Agriculture". The "Agriculture" option is highlighted with a red rectangular border. The main area of the interface is a world map showing various countries and oceans. On the right side of the map, there are navigation controls including a back arrow, zoom in (+) and zoom out (-) buttons, a search icon, and a download icon.

This option is not yet available

CAVA: Other information



This option allows you to select the base maps

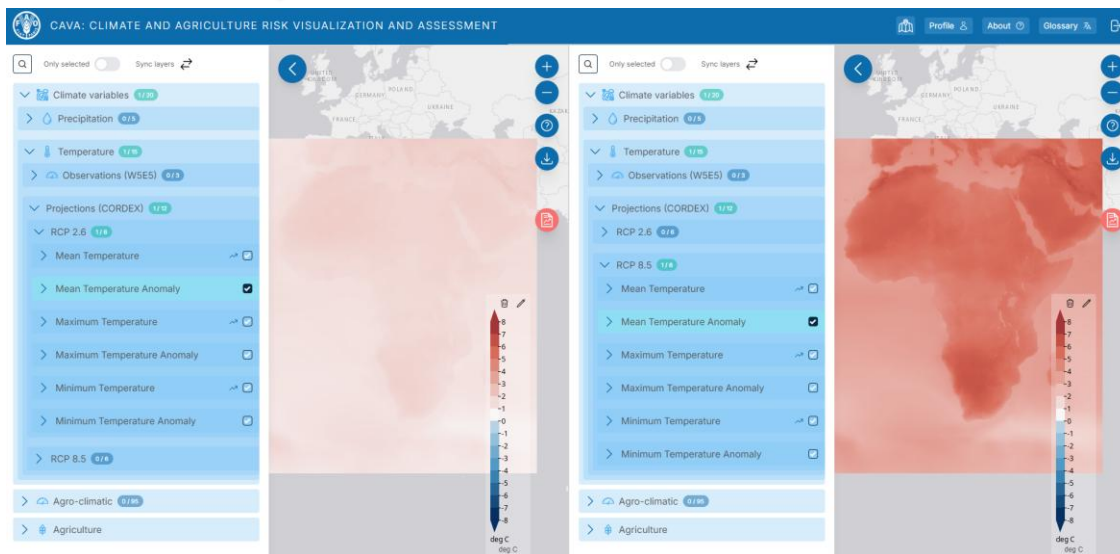
Elevation



Satellite



Hybrid



This option allows you to split the screen in two or four to better compare the results

CAVA: Other information

Glossary

- <

A
- Average amount of rain in rainy days in Agronomic Season
- Agronomic Season length
- Agronomic Season end date
- Average daily tas in Agronomic Season
- >

F
- >

N
- >

S
- >

T

⚠ THIS PAGE IS UNDER DEVELOPMENT AND REVISION ⚠

AVERAGE AMOUNT OF RAIN IN RAINY DAYS IN AGRONOMIC SEASON

The agronomic season is determined in each year as the days between the final season start and ends dates, as determined by soil moisture.

CAVA: Other information

 CAVA: CLIMATE AND AGRICULTURE RISK VISUALIZATION AND ASSESSMENT

[Map](#) [About](#) [Glossary](#)

Climate and Agriculture Risk Visualization and Assessment (CAVA) is an approach to climate information with a focus on agriculture that aims at democratising access and usability of past and future climate (CORDEX) and impacts (ISIMIP) information for the provision of long-term climate services.

CAVA is conceptually organised into two modules.

- Climate module: Provision of climate information from regional climate model outputs (CORDEX-CORE and reanalyses datasets)
- Agriculture module: Provision of impact information from the Inter-Sectoral Impact Model Intercomparison Project (ISIMIP3)

CAVA is a global product but regional CAVA applications can be developed upon request. Regional CAVA applications allow countries to promote the usage of regionally developed climate and impact models.

Climate variables

Observed

Modelled

Agro-climatic

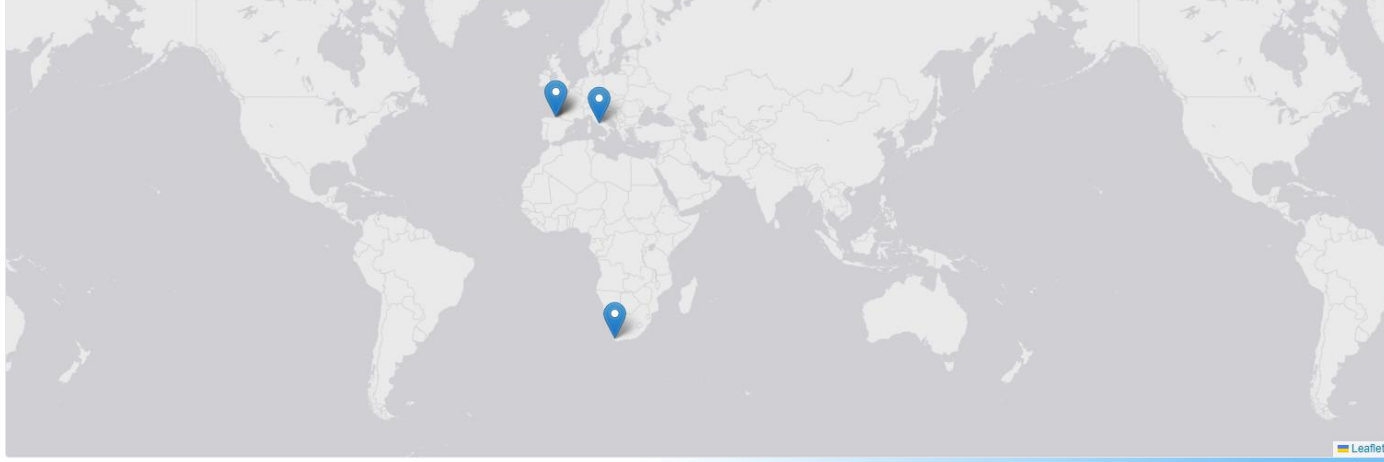
Computed

Agricultural production

Prototype (In progress)

Regional CAVA product

Collected (In progress, upon request)



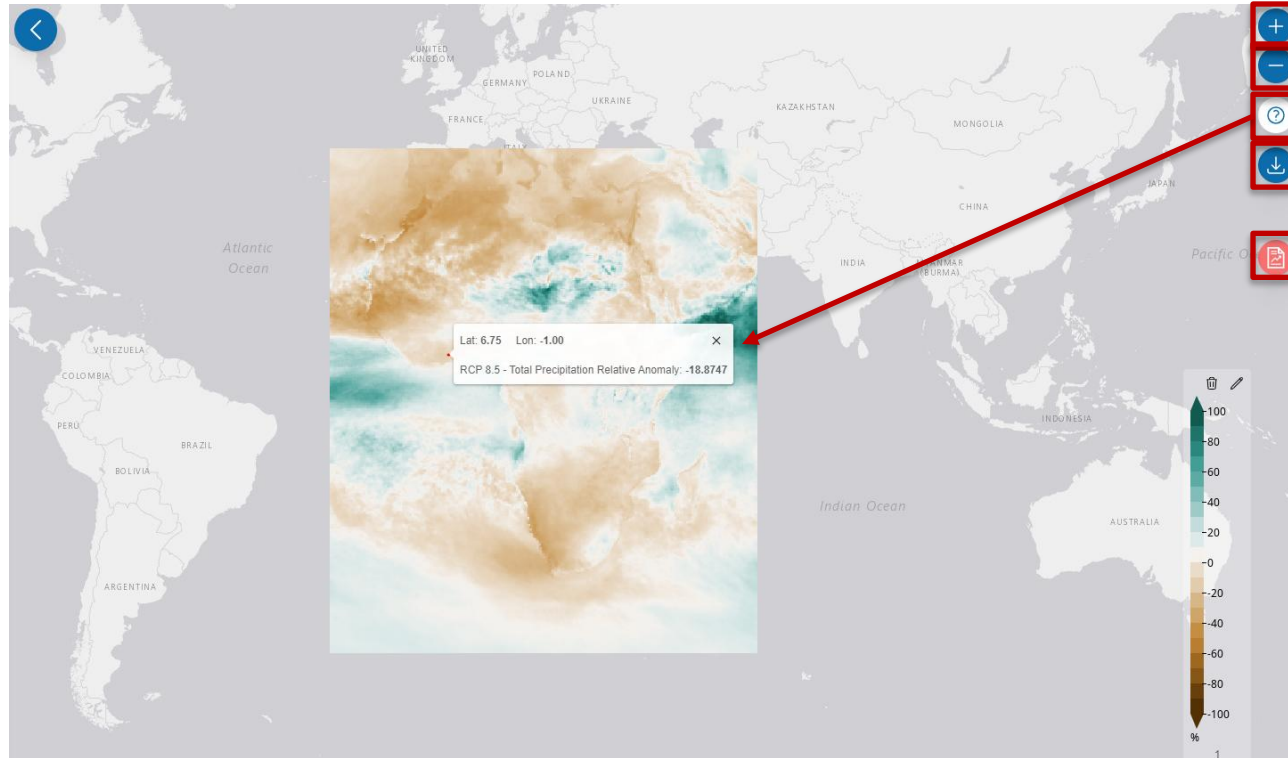
Leaflet

 Food and Agriculture Organization of the United Nations

 Universidad de Cantabria

[Contact](#) [News](#) [Terms and conditions](#)

CAVA: Other information



Zoom-in



Zoom-out



Information for a specific point



Download the map in png format.



Automatic report generation!

Other functionalities: automatic reports

CAVA: Other functionalities (automatic reports)

Step 2: Select the continent of interest (**Asia**)

Step 3: Select the country of interest (**Yemen**)

Step 4: Select the administrative boundaries

You can select second-level boundaries (prefectures) and multiple regions/prefectures at the same time.

Regional information

Asia

Yemen

Admin regions

Al Hudaydah

Report

Daily data

Charts

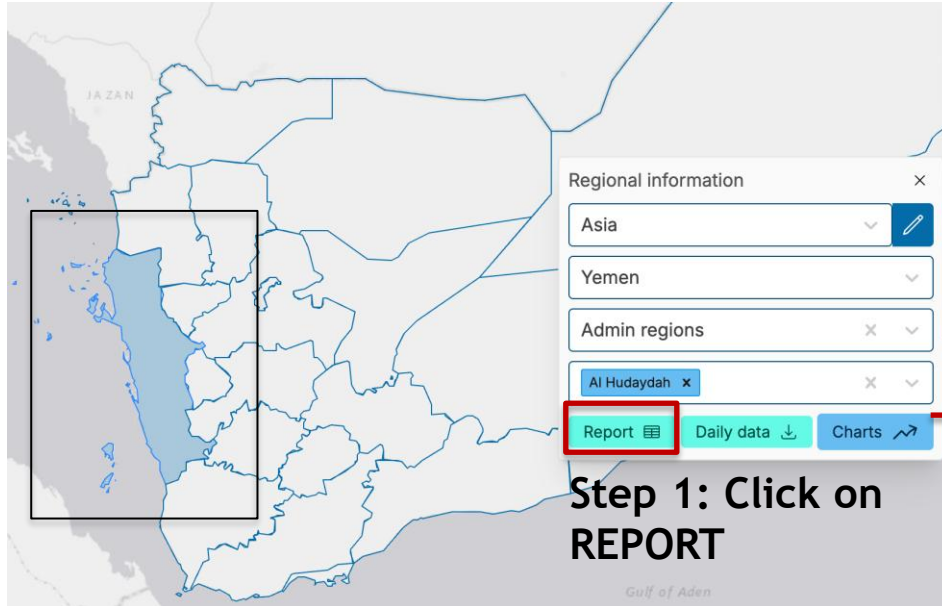
Download a
report

Download
daily data

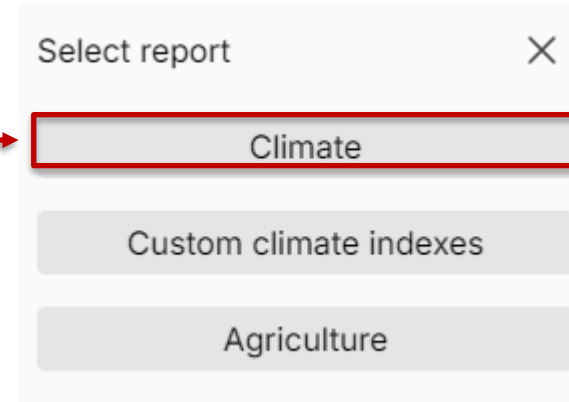
View the
information

Step 1: Click on this
icon

CAVA: Other functionalities (automatic reports)



Here you will see the different types of reports that you can download in CAVA



Step 2: Click on CLIMATE

CAVA will perform an average (set) of all pixels in the Al Hudaydah governorate

You can also add other regions by clicking directly on the map.

CAVA: Other functionalities (automatic reports)

← Fill Climate report

CLIMATE REPORT

The climate report offers ready-to-use figures, insightful explanations, and trend analyses of observed and projected climate conditions for the user-defined area. Analyses are conducted by default for the whole century (2010-2099) and for the historical period (1990-2019)

Email*
andrea.setti@fao.org

Region name*

Region WKT*
MULTIPOLYGON(((42.73470067000005 13.604846509000026,42.71752431700003 13.603

Domain*
Domain refers to geographical areas according to CORDEX initiative. Visit [this page](#) for more information
auto

SEASONS

By default, analyses for the whole year are generated. The option below allows users to specify seasons of interest, for example, January-March (1-3). A minimum of two user-defined seasons are required

SEASON

Start month*
1

End month*
12

The climate report provides ready-to-use figures, relevant explanations, and trend analyses of observed and projected climate conditions for the user-defined area. By default, the analyses are performed for the entire century (2010-2099) and for the historical period (1990-2019).

Step 1: Enter your email

Step 2: Enter the name of your region (**Yemen**)

Step 3: Select the area of interest (**WAS-22**)

Step 4a: Select the season of interest (**1-12 = January-December**)

Remove item

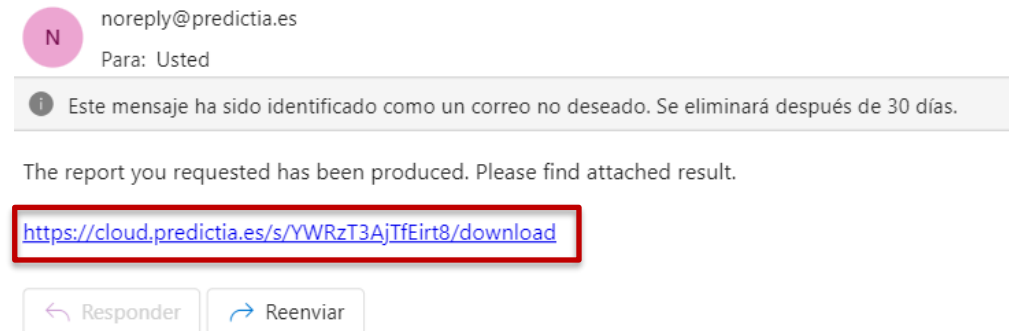
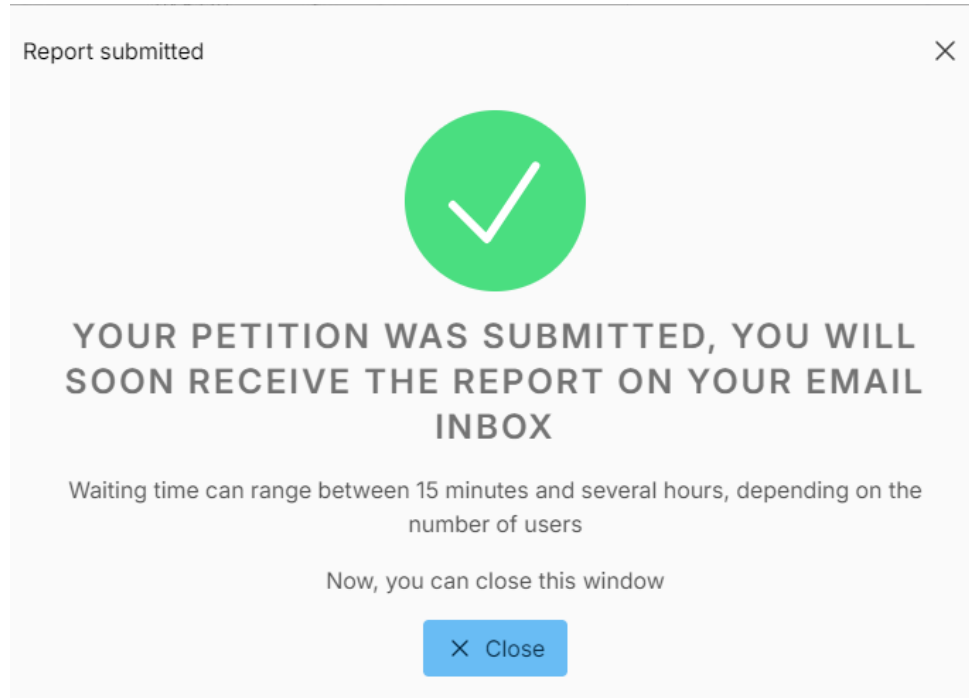
Add item +

Submit

Step 4b: Select another season of interest (**3-7 = March-July**) by clicking **ADD ITEM**

Step 5: Click **SUBMIT** to get the report

CAVA: Other functionalities (automatic reports)



Step 1: Download and unzip the file you received in your email inbox. Check your SPAM folder!!!

2. Interpretation of maps

Interpretation guide

When interpreting graphs, it is important to ensure that the following elements are included.

- The interpretation of the graphs should not exceed three or four sentences - keep it brief and get straight to the point.
- Describe whether the rate of change is statistically significantly different (historical graphs): **pixels showing statistical significance ($p < 0.05$) in quasi-observation plots are marked with a black dot.**
- Describe whether there is agreement on the models for the climate change signal (future graphs): **Crosshatches in future plots indicate a strong agreement (4 models out of 6) of the climate change signal.**
- Feel free to conduct a literature review and compare your results with those presented in national communications, adaptation plans, and/or scientific articles (especially for future graphs).
- The comparison of temporal differences must always be made relative to the reference period (1976-2005) (future graphs).
- Are there spatial differences in the rate of change? If so, what factors are behind these differences (e.g., geographical)? (historical/future graphs)
- Compare the RCPs (future graphs).

Example interpretation: historical climate

Spatial Differences

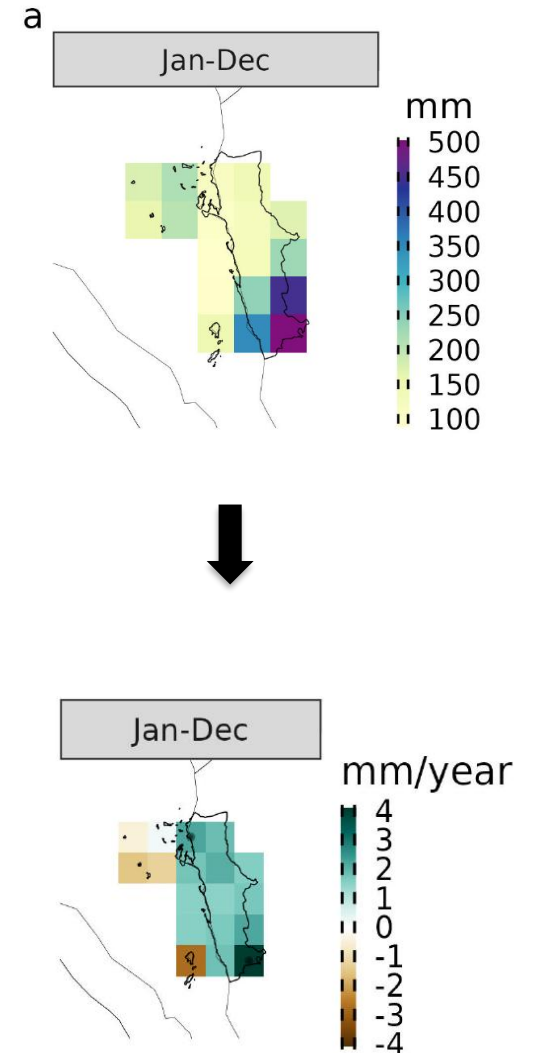
The average total annual rainfall is highest (400-500 mm) in the southern part of the governorate, and lowest (100-150 mm) in the northern parts.

Rate of Change and Spatial Differences

Overall, the governorate experienced a positive anomaly of 2 to 4 mm per year in the average total annual rainfall from 1990 to 2019.

The change is statistically significant in the extreme northern and southern areas of the governorate (pixels showing statistical significance ($p < 0.05$) in quasi-observation plots are marked with a black dot), showing an increasing trend of 4mm/year in total annual rainfall over the same period. This translates to a total increase of 120 mm, equivalent to a 24% increase in total annual rainfall.

Comparison of our results with other research studies



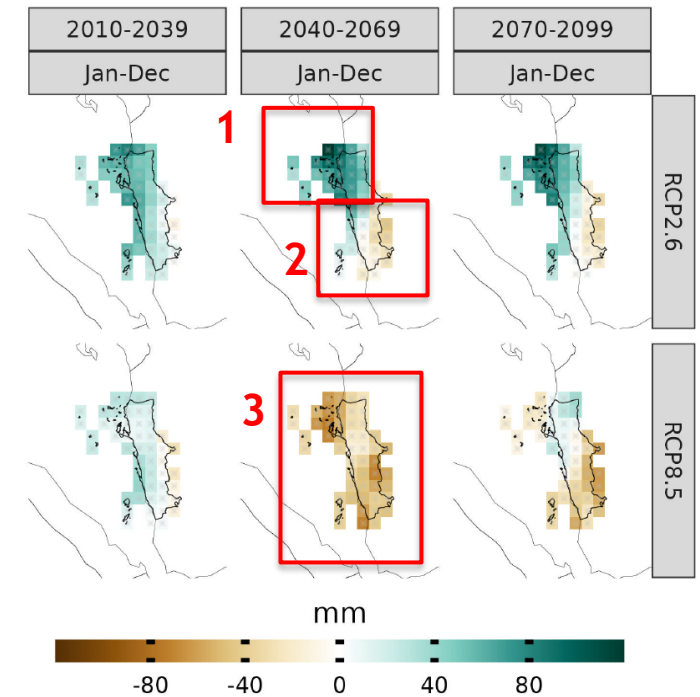
Example interpretation: future climate

Climate change signal

There is strong agreement among models (crosshatches indicating a strong agreement (4 models out of 6) of the climate change signal) regarding changes in total annual precipitation in the Al Hudaydah governorate, both spatially and temporally (crosses)

Rate of Change and Spatial-Temporal Differences

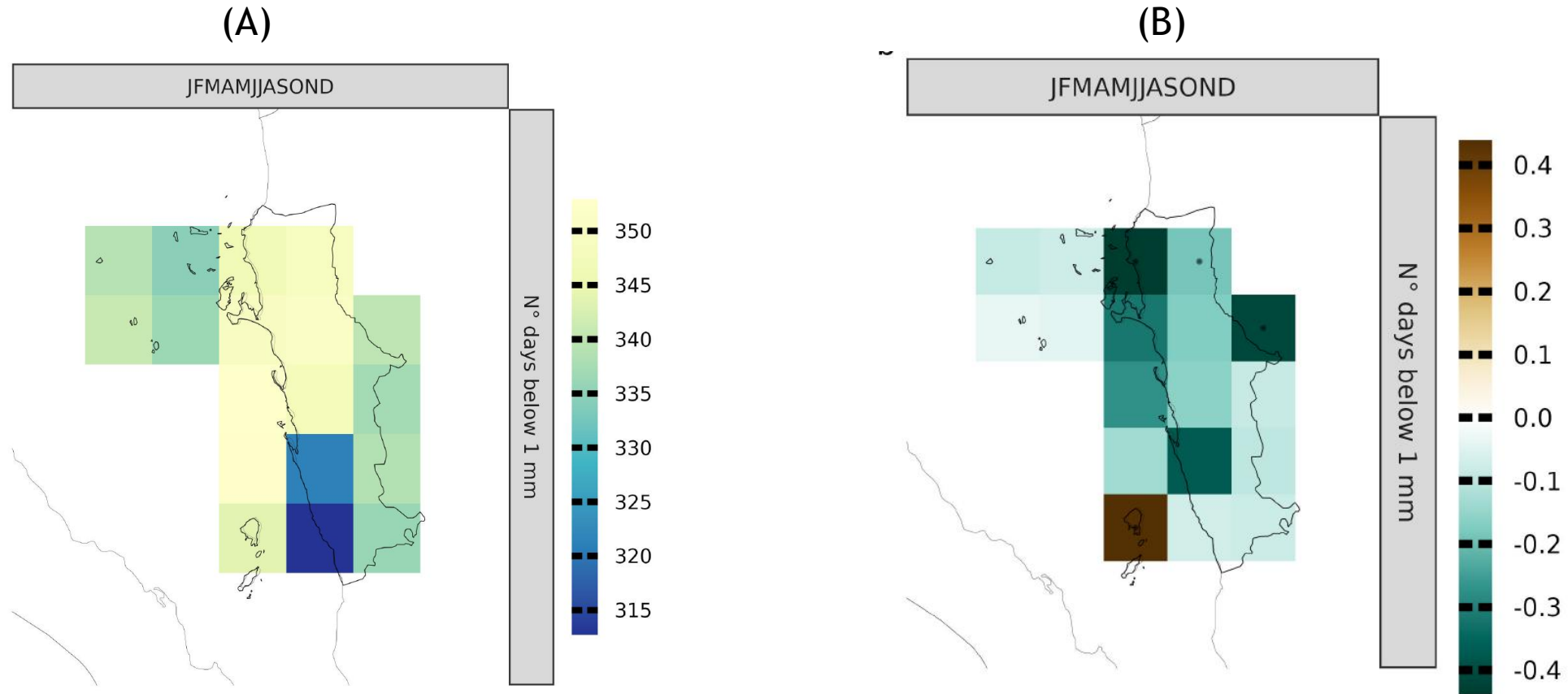
- **RCP 2.6 (low emissions):** Al Hudaydah becomes progressively wetter (+40 to +80mm) in the northern area of the governorate across the century, with increasing positive precipitation anomalies from early to late century. Conversely, the southern area is progressively getting drier from the mid-century (-0 to -40mm).
- **RCP 8.5 (high emissions):** The governorate shifts toward drier conditions, with mild early-century increases (0 to 20mm). Changing into widespread and stronger reductions (-20 to -60mm) in annual rainfall by mid and late century.
- **Overall comparison:** The two scenarios diverge over time. RCP 2.6 leads to increased rainfall except for the southern area, while RCP 8.5 leads to substantial drying, highlighting high sensitivity to global emissions pathways.



Comparison of our results with other research studies

Results automatic report

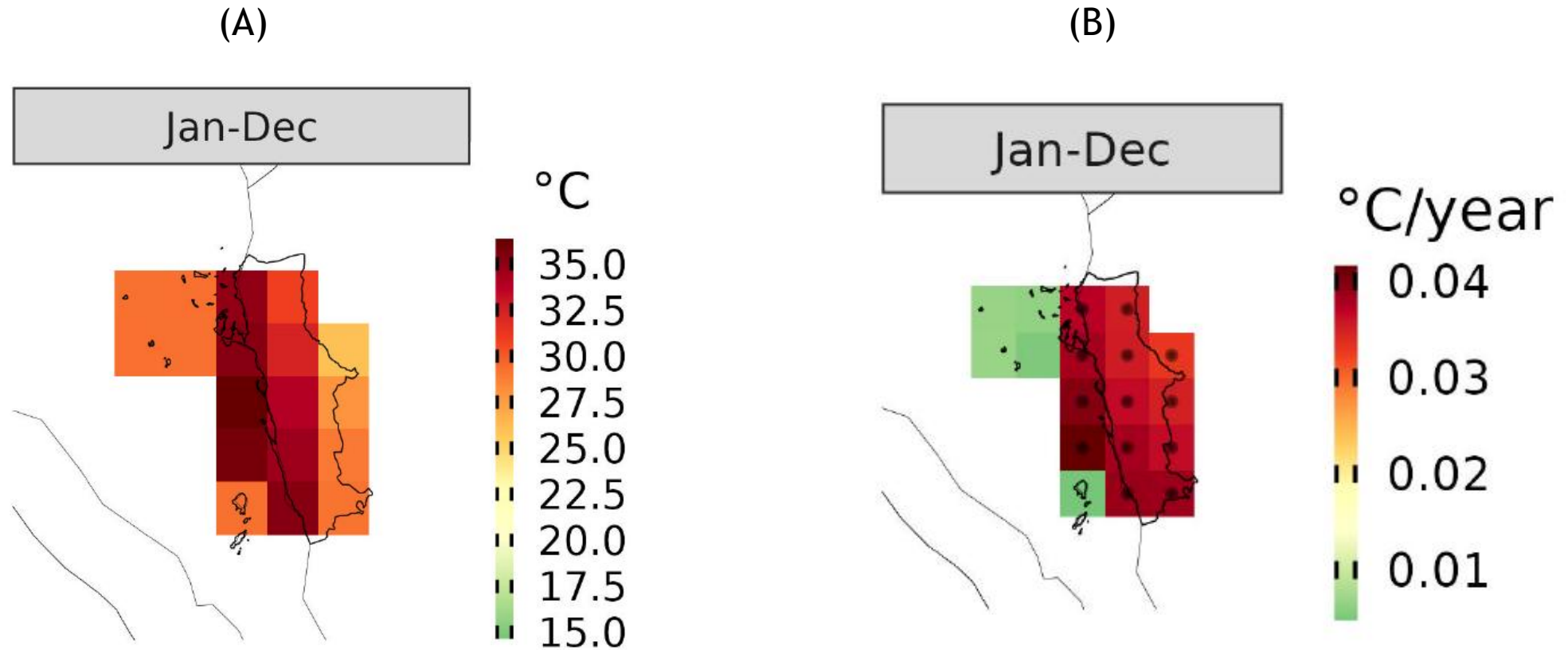
(a) Total annual number of dry days and (b) annual change over the period 1990-2019.



Dry days: annual count of days where precipitation < 1 mm day⁻¹.

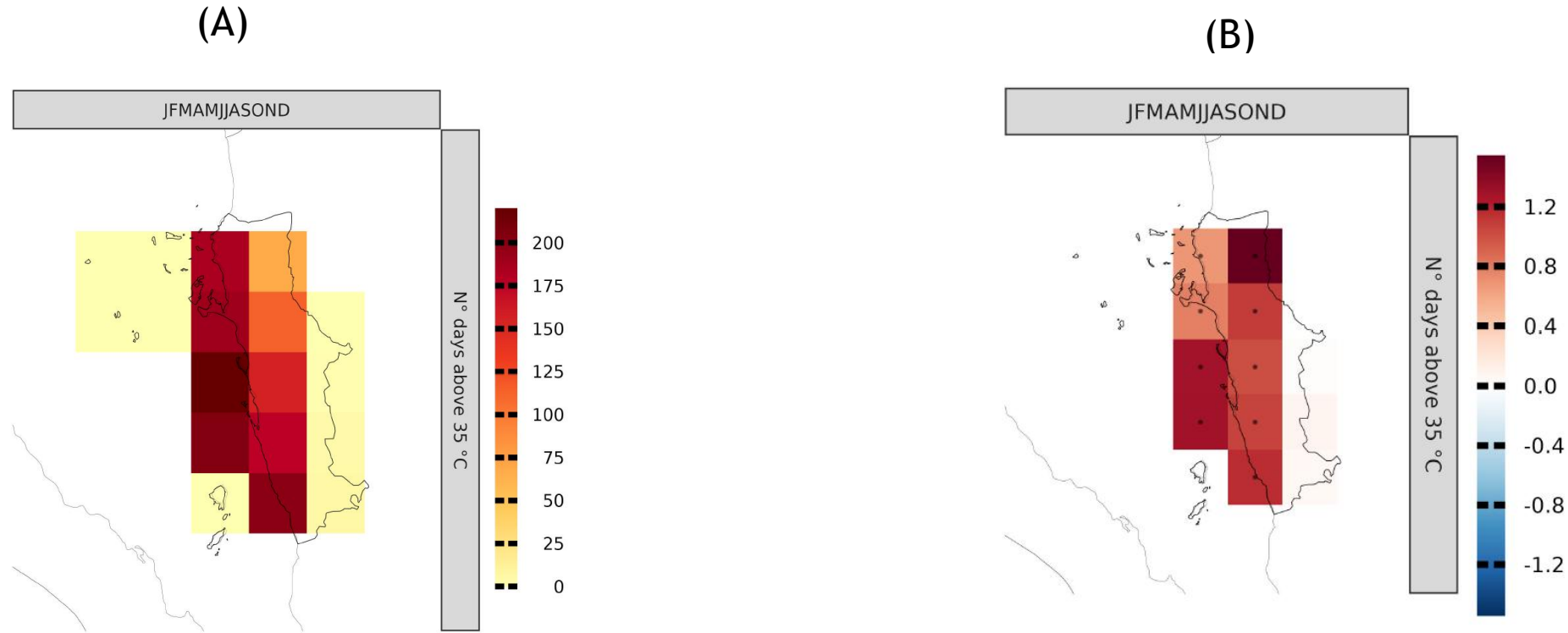
Results automatic report

(a) Average daily maximum temperature and (b) annual change over the period 1990-2019.



Results automatic report

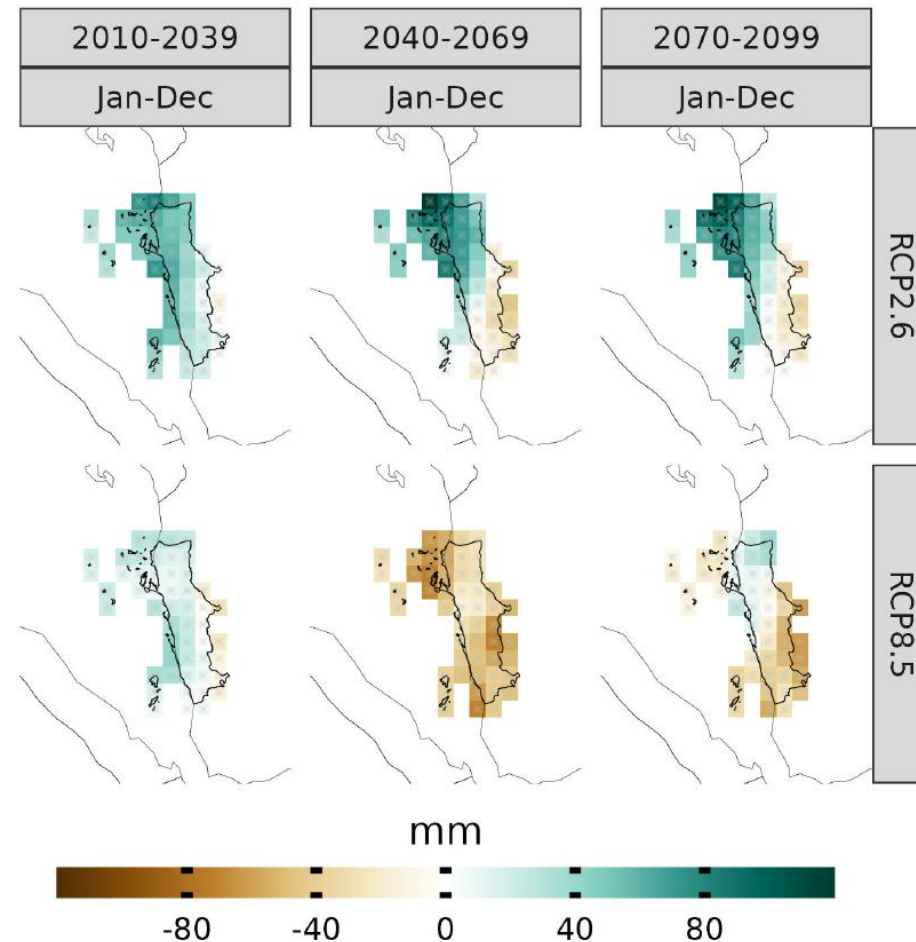
(a) Annual number of hot days and (b) annual change over the period 1990-2019.



Hot days: when TX (daily maximum) > 35 °C

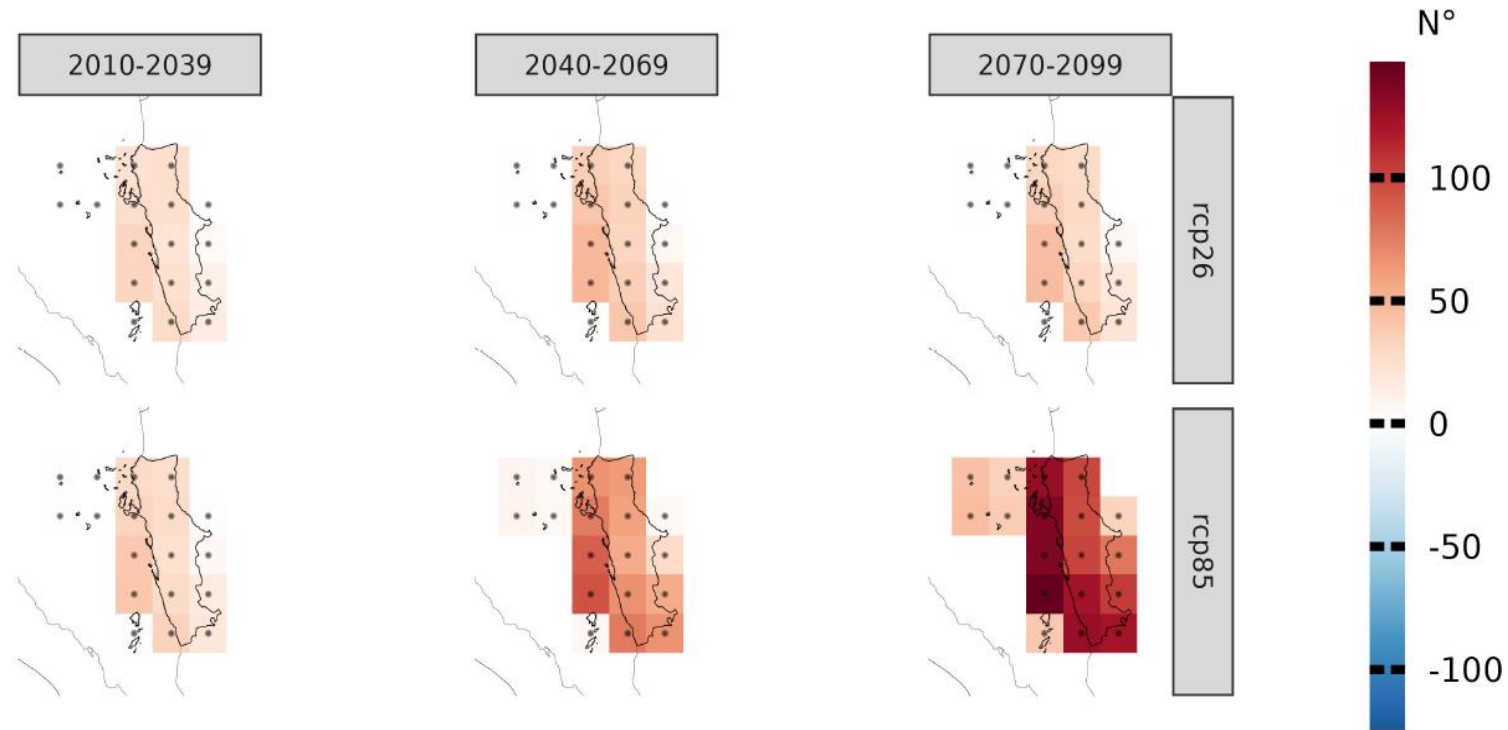
Results automatic report

Change in total mean annual precipitation for RCP 2.6 and 8.5 during the 21st century compared to the period 1976-2005



Results automatic report

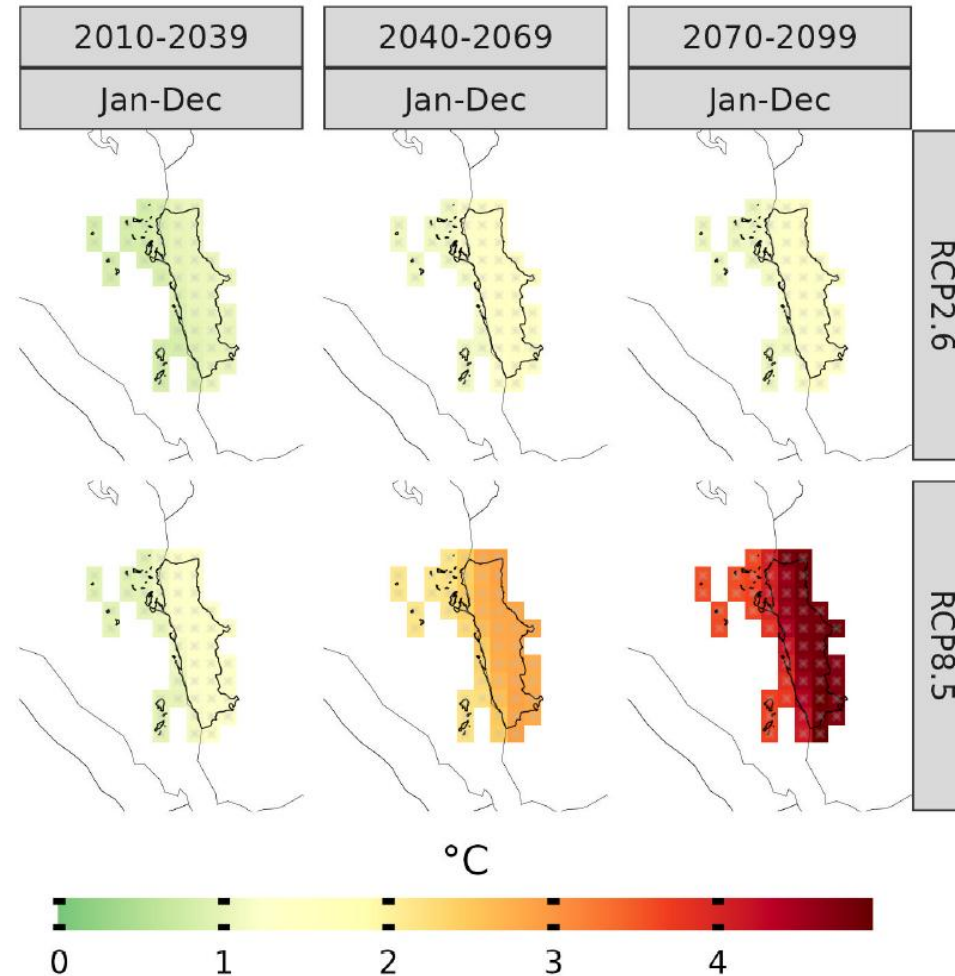
Change in the annual number of hot days for RCP 2.6 and 8.5 during the 21st century compared to the period 1976-2005.



Hot days: when TX (daily maximum) > 35 °C

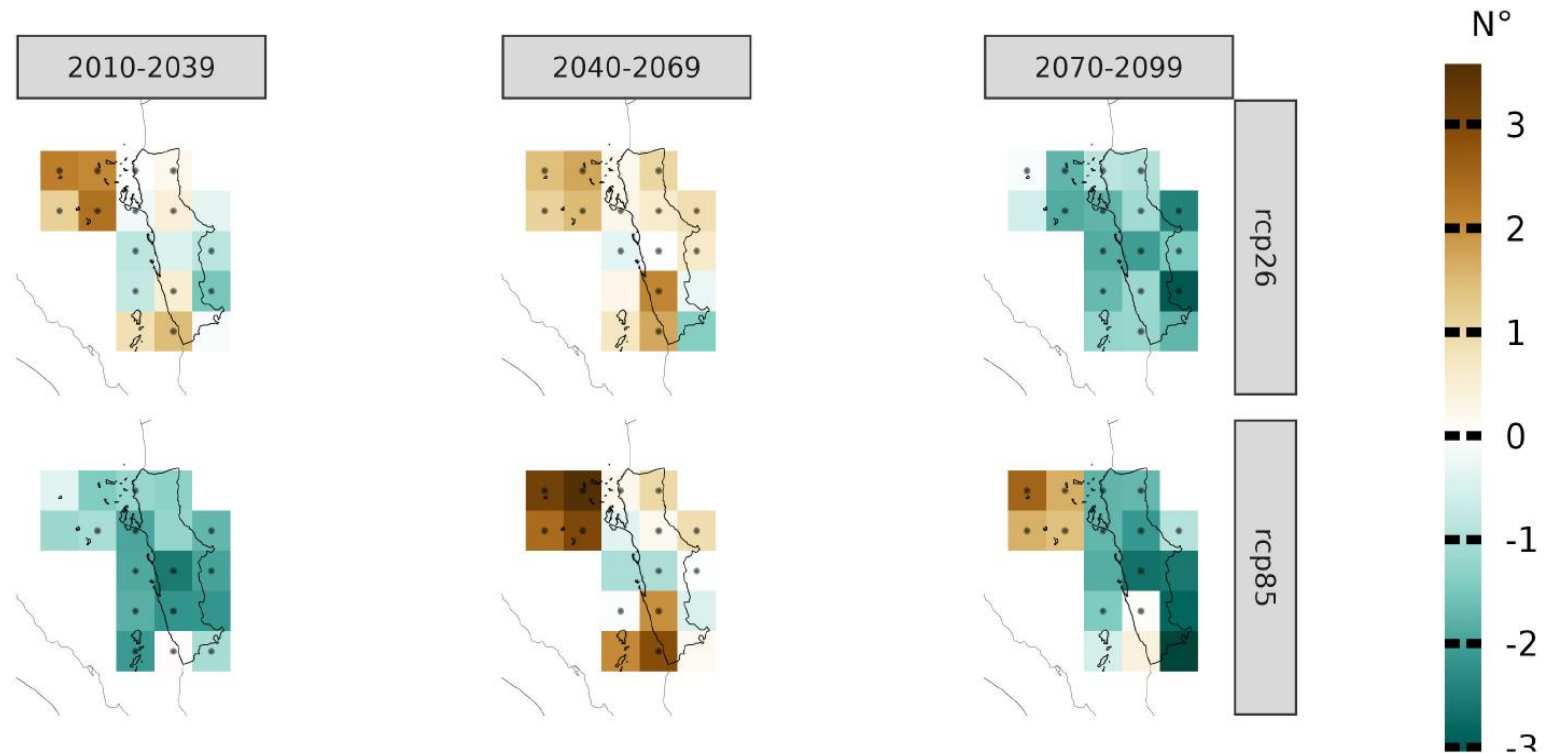
Results automatic report

Change in the average daily maximum temperature for RCP 2.6 and 8.5 during the 21st century compared to the period 1976-2005.



Results automatic report

Change in the annual number of dry days for RCPs 2.6 and 8.5 during the 21st century compared to the period 1976-2005



Dry days: when Precipitation (daily) < 1mm.

End!
